Lecture Notes: Linear Algebra II - 2026

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Constructing New Vector Spaces Out of Old Ones

(Chapter 18)

The Homomorphism Space, The Dual Space And The Dual Map (Section 18.a)

The Homomorphism Space

Proposition 18.a.1 (*The Vector Space Structure of the Homomorphism Space*): Let V and W be two vector spaces over K. Then, $\mathrm{Hom}(V, W)$ has the structure of a vector space over K, if we endow it with the following operations:

- For all $T_1, T_2 \in \mathrm{Hom}(V, W)$, $(T_1 + T_2)(v) := T_1(v) + T_2(v)$ for all $v \in V$.
- For all $T \in \mathrm{Hom}(V, W)$ and $\alpha \in K$, $(\alpha \cdot T)(v) := \alpha \cdot T(v)$ for all $v \in V$.

Proof: We claim that $T_1 + T_2 \in \mathrm{Hom}(V, W)$, meaning $T_1 + T_2$ is a linear map. Indeed, for a scalar a and a vector v, we have

$$\begin{aligned}(T_1 + T_2)(a \cdot v) &= T_1(a \cdot v) + T_2(a \cdot v) = a \cdot T_1(v) + a \cdot T_2(v) \\ &= a \cdot (T_1(v) + T_2(v)) = a \cdot (T_1 + T_2)(v),\end{aligned}$$

and for vectors v_1, v_2, we have

$$\begin{aligned}(T_1 + T_2)(v_1 + v_2) &= T_1(v_1 + v_2) + T_2(v_1 + v_2) = T_1(v_1) + T_1(v_2) + T_2(v_1) + T_2(v_2) \\ &= (T_1 + T_2)(v_1) + (T_1 + T_2)(v_2).\end{aligned}$$

Consequently, $T_1 + T_2$ is a linear map. The proof that $\alpha \cdot T$ is linear is analogous.

So far, we have defined $+\colon \mathrm{Hom}(V, W) \times \mathrm{Hom}(V, W) \to \mathrm{Hom}(V, W)$ and $\cdot\colon K \times \mathrm{Hom}(V, W) \to \mathrm{Hom}(V, W)$.

Regarding the neutral element, the map $\mathcal{O}\colon V \to W$ defined by $v \mapsto 0$ for all $v \in V$ is linear. We claim that $\mathcal{O} + T = T$. Indeed,

$$(\mathcal{O} + T)(v) = \mathcal{O}(v) + T(v) = 0 + T(v) = T(v)$$

for all $v \in V$. Similarly, one shows that $T + \mathcal{O} = T$. ■

Exercise: Complete the proof that $\mathrm{Hom}(V, W)$ is a vector space by checking that it satisfies all the axioms of a vector space.

For example, consider the distributivity axiom. For all $\alpha \in K$ and $T_1, T_2 \in \mathrm{Hom}(V, W)$, we have $\alpha \cdot (T_1 + T_2) = \alpha \cdot T_1 + \alpha \cdot T_2$. Indeed, for $v \in V$,

$$\begin{aligned}(\alpha \cdot (T_1 + T_2))(v) &= \alpha \cdot (T_1 + T_2)(v) = \alpha(T_1(v) + T_2(v)) \\ &= \alpha \cdot T_1(v) + \alpha \cdot T_2(v) = (\alpha \cdot T_1)(v) + (\alpha \cdot T_2)(v) \\ &= (\alpha \cdot T_1 + \alpha \cdot T_2)(v).\end{aligned}$$

Therefore, $\alpha \cdot (T_1 + T_2) = \alpha \cdot T_1 + \alpha \cdot T_2$.

Theorem 18.a.2 (*The Canonical Isomorphism to Matrix Space*): Let V and W be finite-dimensional vector spaces over K, let $\mathcal{B} = (v_1, \ldots, v_n)$ be a basis for V, and let $\mathcal{C} = (w_1, \ldots, w_m)$ be a basis for W. Then, $\Psi_{\mathcal{C}}^{\mathcal{B}}\colon \mathrm{Hom}(V, W) \to M_{m \times n}(K)$, defined by

$$\Psi_{\mathcal{C}}^{\mathcal{B}}(T) := [T]_{\mathcal{C}}^{\mathcal{B}} = \begin{pmatrix} | & & | \\ [T(v_1)]_{\mathcal{C}} & \cdots & [T(v_n)]_{\mathcal{C}} \\ | & & | \end{pmatrix},$$

is a linear map, and moreover, an isomorphism.

Proof: Recall that $T_{[T]_{\mathcal{C}}^{\mathcal{B}}} = \Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1}$. Recall also that $\Phi_{\mathcal{B}}(v) = [v]_{\mathcal{B}} \in K^n$ for all $v \in V$. Furthermore, $\Phi_{\mathcal{C}}(w) = [w]_{\mathcal{C}} \in K^m$ for all $w \in W$. We have

$$\Phi_{\mathcal{B}}^{-1}\begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} = a_1 v_1 + \cdots + a_n v_n, \quad \Phi_{\mathcal{C}}^{-1}\begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix} = b_1 w_1 + \cdots + b_m w_m.$$

We claim that the map $\Psi_{\mathcal{C}}^{\mathcal{B}}$ is bijective. Indeed, define $\Theta\colon M_{m\times n}(K) \to \mathrm{Hom}(V,W)$ by $\Theta(A) := \Phi_{\mathcal{C}}^{-1} \circ T_A \circ \Phi_{\mathcal{B}} \in \mathrm{Hom}(V,W)$ for $A \in M_{m\times n}(K)$. We have:

$$\begin{aligned}
\Psi_{\mathcal{C}}^{\mathcal{B}} \circ \Theta(A) &= \Psi_{\mathcal{C}}^{\mathcal{B}}(\Phi_{\mathcal{C}}^{-1} \circ T_A \circ \Phi_{\mathcal{B}}) \\
&= \big([\Phi_{\mathcal{C}}^{-1} \circ T_A \circ \Phi_{\mathcal{B}}(v_1)]_{\mathcal{C}} \ldots [\Phi_{\mathcal{C}}^{-1} \circ T_A \circ \Phi_{\mathcal{B}}(v_n)]_{\mathcal{C}}\big) \\
&= \begin{pmatrix} | & & | \\ T_A\Phi_{\mathcal{B}}(v_1) & \ldots & T_A\Phi_{\mathcal{B}}(v_n) \\ | & & | \end{pmatrix} \\
&= \begin{pmatrix} | & & | \\ T_A(e_1) & \ldots & T_A(e_n) \\ | & & | \end{pmatrix} = A.
\end{aligned}$$

This uses the facts that $[\Phi_{\mathcal{C}}^{-1}(b)]_{\mathcal{C}} = b$ for all $b \in K^m$, and $\Phi_{\mathcal{B}}(v_k) = e_k$. This proves that $\Psi_{\mathcal{C}}^{\mathcal{B}} \circ \Theta = \mathrm{id}_{M_{m\times n}(K)}$.

Now, by definition, $\Theta \circ \Psi_{\mathcal{C}}^{\mathcal{B}}(T) = \Phi_{\mathcal{C}}^{-1} \circ T_{\Psi_{\mathcal{C}}^{\mathcal{B}}(T)} \circ \Phi_{\mathcal{B}} = \Phi_{\mathcal{C}}^{-1} \circ T_{[T]_{\mathcal{C}}^{\mathcal{B}}} \circ \Phi_{\mathcal{B}} = T$. This shows that $\Theta \circ \Psi_{\mathcal{C}}^{\mathcal{B}} = \mathrm{id}_{\mathrm{Hom}(V,W)}$. Therefore, $\Psi_{\mathcal{C}}^{\mathcal{B}}$ is bijective.

It remains to prove that $\Psi_{\mathcal{C}}^{\mathcal{B}}$ is linear. Indeed, let $\alpha \in K$, and let $T \in \mathrm{Hom}(V,W)$. Then,

$$\begin{aligned}
\Psi_{\mathcal{C}}^{\mathcal{B}}(\alpha \cdot T) &= ([(\alpha T)(v_1)]_{\mathcal{C}} \ldots [(\alpha T)(v_n)]_{\mathcal{C}}) \\
&= ([\alpha T(v_1)]_{\mathcal{C}} \ldots [\alpha T(v_n)]_{\mathcal{C}}) \\
&= (\alpha[T(v_1)]_{\mathcal{C}} \ldots \alpha[T(v_n)]_{\mathcal{C}}) \\
&= \alpha \cdot \begin{pmatrix} | & & | \\ [T(v_1)]_{\mathcal{C}} & \ldots & [T(v_n)]_{\mathcal{C}} \\ | & & | \end{pmatrix} \\
&= \alpha \cdot \Psi_{\mathcal{C}}^{\mathcal{B}}(T).
\end{aligned}$$

Similarly, one shows that $\Psi_{\mathcal{C}}^{\mathcal{B}}(T_1 + T_2) = \Psi_{\mathcal{C}}^{\mathcal{B}}(T_1) + \Psi_{\mathcal{C}}^{\mathcal{B}}(T_2)$. ■

Corollary 18.a.3 (*Dimensionality of the Homomorphism Space*): If V and W are finite-dimensional vector spaces, then so is $\mathrm{Hom}(V,W)$, and $\dim \mathrm{Hom}(V,W) = \dim V \cdot \dim W$.

Proof: Put $n := \dim V$ and $m := \dim W$. By the previous theorem, we have $\mathrm{Hom}(V,W) \cong M_{m\times n}(K)$, which implies that $\mathrm{Hom}(V,W)$ is finite-dimensional. Moreover, $\dim \mathrm{Hom}(V,W) = \dim M_{m\times n}(K) = m \cdot n$. ■

Rings

Example: **(a)** Every field K is also a commutative ring.

(b) $\mathbb{Z}$ is a commutative ring.

(c) $M_{n\times n}(K)$ is a ring, where $A \cdot B$ is matrix multiplication, and the unity is I_n. Clearly, it is not commutative.

Corollary 18.a.4 (*The Ring of Endomorphisms*): Let V be a vector space over K. Then, $\mathrm{End}(V) := \mathrm{Hom}(V,V)$ is a ring with a unity. This ring is, in general, not commutative. The multiplication in $\mathrm{End}(V)$ is given by composition $T_1 \cdot T_2 := T_1 \circ T_2$ for all $T_1, T_2 \in \mathrm{Hom}(V,V)$, and the unity is id_V. Moreover, if V is finite-dimensional with $\dim V = n$, and $\mathcal{B} = (w_1, \ldots, w_n)$ is a basis for V, then $\Psi_{\mathcal{B}}^{\mathcal{B}}\colon \mathrm{Hom}(V,V) \to M_{n\times n}(K)$ is an isomorphism of rings.

Proof (sketch): The multiplication $T_1 \cdot T_2 := T_1 \circ T_2$ is associative, since $(T_1 \cdot T_2) \cdot T_3 = (T_1 \circ T_2) \circ T_3 = T_1 \circ (T_2 \circ T_3) = T_1 \cdot (T_2 \cdot T_3)$. Also, $T \cdot \mathrm{id}_V = T \circ \mathrm{id}_V = T$, and $\mathrm{id}_V \cdot T = \mathrm{id}_V \circ T = T$. Furthermore, $T \cdot (T_1 + T_2) = T \circ (T_1 + T_2) = T \circ T_1 + T \circ T_2$.

Now, if V is finite-dimensional, and $\mathcal{B}$ is a basis for V, then

$$\Psi_{\mathcal{B}}^{\mathcal{B}}(T_1 \circ T_2) = [T_1 \circ T_2]_{\mathcal{B}}^{\mathcal{B}} = [T_1]_{\mathcal{B}}^{\mathcal{B}} \cdot [T_2]_{\mathcal{B}}^{\mathcal{B}} = \Psi_{\mathcal{B}}^{\mathcal{B}}(T_1) \cdot \Psi_{\mathcal{B}}^{\mathcal{B}}(T_2)$$

via matrix multiplication. ■

The Dual Space

An important special case of $\mathrm{Hom}(V, W)$ is when $W = K$.

Definition 18.a.5 (*The Dual Space and Linear Functionals*): Let V be a vector space over K. The dual space of V is defined to be the vector space $V^* := \mathrm{Hom}(V, K)$. The elements of V^* are linear maps $l \colon V \to K$, and are sometimes called functionals.

Corollary 18.a.6 (*Dimensionality of the Dual Space*): Let V be a finite-dimensional vector space over K. Then, $\dim V^* = \dim V$.

Proof: We have $\dim V^* = \dim \mathrm{Hom}(V, K) = \dim V \cdot \dim K = \dim V \cdot 1 = \dim V$. ■

Let $V = K_{\mathrm{col}}^n$. Then, we can identify V^* with K_{row}^n as follows: If $l \in K_{\mathrm{row}}^n$, then l defines a functional $f_l \colon K_{\mathrm{col}}^n \to K$ by $f_l(v) := l \cdot v \in M_{1\times 1}(K) \cong K$. The map $D \colon K_{\mathrm{row}}^n \to (K_{\mathrm{col}}^n)^*$ defined by $D(l) := f_l$ is a linear map, and therefore, it is an isomorphism.

Let V be a finite-dimensional vector space over K, and let $\mathcal{B} = (v_1, \ldots, v_n)$ be a basis for V. For $1 \le i \le n$, define a functional $v_i^* \in V^*$ as follows: $v_i^*(v_j) := \delta_{ij}$. This means that $v_i^*(v_i) = 1$, and $v_i^*(v_j) = 0$ for all $j \neq i$. Since $v_1, \ldots, v_n$ form a basis for V, our definition determines uniquely a well-defined functional $v_i^* \colon V \to K$.

Lemma 18.a.7 (*Construction of the Dual Basis*): The elements $v_1^*, \ldots, v_n^* \in V^*$ form a basis for V^*. This basis is called the dual basis to $\mathcal{B}$. We denote this basis by $\mathcal{B}^* = (v_1^*, \ldots, v_n^*)$. Moreover, for all $l \in V^*$, $l = \sum_{i=1}^n l(v_i) \cdot v_i^*$.

Proof: Let $l \in V^*$. We claim that $l = \sum_{i=1}^n l(v_i) \cdot v_i^*$, or in other words, we claim that $l(v) = \sum_{i=1}^n l(v_i) \cdot v_i^*(v)$. Indeed, both sides of the equation are linear maps $V \to K$, and therefore, it is enough to check that it holds for $v = v_1, v = v_2, \ldots, v = v_n$, because $v_1, \ldots, v_n$ form a basis for V. And indeed, for $v = v_k$, we have:

$$\sum_{i=1}^n l(v_i) \cdot v_i^*(v_k) = \sum_{i=1}^n l(v_i) \cdot \delta_{ik} = l(v_k).$$

The claim just proven shows that $v_1^*, \ldots, v_n^*$ span V^*. Since $\dim V^* = \dim V = n$, it follows that $v_1^*, \ldots, v_n^*$ form a basis for V^*. ■

Question: Let V be a finite-dimensional vector space over K, and let $\mathcal{B}$ and $\mathcal{C}$ be two bases for V. $\mathcal{B}^*$ and $\mathcal{C}^*$ are two bases for V^*. What is the relation between $[\mathrm{id}_{V^*}]_{\mathcal{B}^*}^{\mathcal{C}^*}$ and $[\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{C}}$?

Proposition 18.a.8 (*Dual Change of Basis and Transposition*): The relationship between transition matrices in the dual space and the primal space is characterized by transposition and inversion. Specifically, the transition matrix between dual bases $\mathcal{C}^*$ and $\mathcal{B}^*$ is the transpose of the inverse transition matrix between the original bases $\mathcal{C}$ and $\mathcal{B}$:

$$[\mathrm{id}_{V^*}]_{\mathcal{B}^*}^{\mathcal{C}^*} = \left([\mathrm{id}_V]_{\mathcal{C}}^{\mathcal{B}}\right)^T = \left(\left([\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{C}}\right)^{-1}\right)^T.$$

Thus, the coordinate transformation in the dual space corresponds to the transpose of the inverse transformation in the primal space.

Proof: Let $\mathcal{B} = (v_1, \ldots, v_n)$, and let $\mathcal{C} = (w_1, \ldots, w_n)$. We have $A := [\mathrm{id}_V]_{\mathcal{C}}^{\mathcal{B}} = \begin{pmatrix} | & & | \\ [v_1]_{\mathcal{C}} & \ldots & [v_n]_{\mathcal{C}} \\ | & & | \end{pmatrix}$.

Write $A = (a_{ij})$. By definition, $v_i = \sum_{k=1}^n a_{ki} w_k$. Let $\mathcal{C}^* = (w_1^*, \ldots, w_n^*)$ be the dual basis of $\mathcal{C}$. We apply the previous lemma to $l = w_j^*$ and the basis $\mathcal{B}$:

$$w_j^* = \sum_{i=1}^n w_j^*(v_i) \cdot v_i^* = \sum_{i=1}^n w_j^* \left(\sum_{k=1}^n a_{ki} w_k \right) \cdot v_i^* = \sum_{i=1}^n a_{ji} v_i^*.$$

Here, only $k = j$ remains from the inner sum. This implies that $[\mathrm{id}_{V^*}]_{\mathcal{B}^*}^{\mathcal{C}^*} = A^T = ([\mathrm{id}_V]_{\mathcal{C}}^{\mathcal{B}})^T$. ■

The Dual Map

Lemma 18.a.9 (*Definition of the Dual Map*): Let V and W be vector spaces over K, and let $T\colon V \to W$ be a linear map. Define a new map $T^*\colon W^* \to V^*$ by $T^*(l) := l \circ T$ for all $l \in W^*$. Then, T^* is a linear map. It is called the dual map to T.

Proof: Let $l \in W^*$, meaning $l\colon W \to K$ is linear. Clearly, $l \circ T\colon V \to K$ is a linear map because both T and l are linear, so $T^*(l) := l \circ T \in V^*$. This shows that $T^*\colon W^* \to V^*$ has its target indeed in V^*, as claimed.

We will show now that T^* is a linear map. Let $l \in W^*$, and let $\alpha \in K$. We have that $T^*(\alpha l) \in V^*$ is the map:

$$T^*(\alpha l)(v) = ((\alpha l) \circ T)(v) = (\alpha l)(T(v)) = \alpha l(T(v)) = \alpha \cdot (l \circ T)(v) = \alpha(T^* l)(v).$$

So, $T^*(\alpha l) = \alpha \cdot T^*(l)$. Similarly, one shows that $T^*(l_1 + l_2) = T^*(l_1) + T^*(l_2)$. ■

Proposition 18.a.10 (*Functoriality and Composition of Duals*): Let U, V, W be vector spaces over K, and let $T\colon V \to W$ and $S\colon W \to U$ be linear maps. Consider the dual maps $T^*\colon W^* \to V^*$ and $S^*\colon U^* \to W^*$. Then, $(S \circ T)^* = T^* \circ S^*$.

Proof: Exercise. ■

Exercise: Let V be a vector space over K. Show that:

(a) $(\mathrm{id}_V)^* = \mathrm{id}_{V^*}$.

(b) $(\mathcal{O}\colon V \to V)^* = (\mathcal{O}\colon V^* \to V^*)$.

In summary, applying $(-)^*$, or $\mathrm{Hom}(-, K)$, takes every vector space V and gives us a new vector space V^*. It also takes any linear map $T\colon V \to W$ and gives us a new map T^* between the duals of V and W, but the arrow is reversed: $T^*\colon W^* \to V^*$. This is an example of a contravariant functor.

Lemma 18.a.11 (*Transposition of the Representation Matrix*): Let $S\colon V \to W$ be a linear map between two finite-dimensional vector spaces V and W over K. Let $\mathcal{B}$ be a basis for V, and let $\mathcal{C}$ be a basis for W. Consider $S^*\colon W^* \to V^*$ and the bases $\mathcal{C}^*, \mathcal{B}^*$ for W^*, V^*, respectively. Then, $[S^*]_{\mathcal{B}^*}^{\mathcal{C}^*} = ([S]_{\mathcal{C}}^{\mathcal{B}})^T$.

Proof: Write $\mathcal{B} = (v_1, \ldots, v_n)$, $\mathcal{C} = (w_1, \ldots, w_m)$, and $A := [S]_{\mathcal{C}}^{\mathcal{B}} = (a_{ij})$. A satisfies $S(v_j) = \sum_{i=1}^m a_{ij} w_i$. Now, for $w_j^* \in \mathcal{C}^*$, we have

$$S^*(w_j^*) = w_j^* \circ S = \sum_{i=1}^n (w_j^* \circ S)(v_i) \cdot v_i^* = \sum_{i=1}^n w_j^*(S(v_i)) \cdot v_i^*,$$

where the second equality follows from the Lemma on the Dual Basis. Substituting $S(v_i)$, we get

$$= \sum_{i=1}^n w_j^* \left(\sum_{k=1}^m a_{ki} w_k \right) \cdot v_i^* = \sum_{i=1}^n a_{ji} v_i^*.$$

The coordinates of $S^*(w_j^*)$ in the basis $\mathcal{B}^* = (v_1^*, \ldots, v_n^*)$ are $\begin{pmatrix} a_{j1} \\ \vdots \\ a_{jn} \end{pmatrix}$, which corresponds to row j in A, represented as a column (i.e., column j in A^T). ■

Reflexivity

Let V be a vector space over K. We have V^*, but also $V^{**} := (V^*)^*$, obtained by applying the "dual" operation one more time to V^*. V^{**} is sometimes called the bidual space of V. Is there any relation between V^{**} and V?

Theorem 18.a.12 (*The Canonical Isomorphism to the Bidual Space***):** Let V be a finite-dimensional vector space over K. Then, there is a canonical isomorphism $\tau\colon V \to (V^*)^*$ given by: for $v \in V$, $\tau(v)\colon V^* \to K$ is the map $\tau(v)(l) := l(v)$ for all $l \in V^*$.

Remark: The map $\tau\colon V \to (V^*)^*$ exists and is defined in the same way as above regardless of V being finite- or infinite-dimensional. Moreover, τ is always injective. However, when V is infinite-dimensional, τ fails to be surjective.

Proof: We claim that τ is a linear map. Indeed, if $v \in V$ and $\alpha \in K$, we have

$$\begin{aligned}\tau(\alpha v) &= (V^* \ni l \mapsto l(\alpha v) \in K)\\ &= (V^* \ni l \mapsto \alpha l(v) \in K)\\ &= \alpha \cdot (V^* \ni l \mapsto l(v) \in K) = \alpha\tau(v).\end{aligned}$$

Similarly, one shows that $\tau(v' + v'') = \tau(v') + \tau(v'')$.

We now claim that τ is injective. To see this, we will show that $\ker(\tau) = 0$. Indeed, let $v \in \ker(\tau)$, meaning $\tau(v) = 0$. Then, $l(v) = 0$ for all $l \in V^*$. Choose a basis $\mathcal{B} = (v_1, \ldots, v_n)$ for V. Write $v = c_1v_1 + \cdots + c_nv_n$. Apply v_k^* to v, and we obtain $v_k^*(c_1v_1 + \cdots + c_nv_n) = 0$. But $v_k^*(c_1v_1 + \cdots + c_nv_n) = c_k$. Therefore, $c_k = 0$, and this holds for $1 \le k \le n$, which implies that $v = 0$. This proves that τ is injective.

To finish the proof, note that $\dim(V^*)^* = \dim V^* = \dim V$. So, since $\tau\colon V \to (V^*)^*$ is injective, it must be an isomorphism. ■

Remark: The previous theorem is not true for infinite-dimensional vector spaces. More specifically, $\tau\colon V \to (V^*)^*$ is injective, but not surjective. In fact, when V is infinite-dimensional, $(V^*)^*$ is not isomorphic to V.

Naturality

The map $\tau\colon V \to (V^*)^*$ exists for every space V, and it is defined in the same way. To distinguish τ's for different spaces, we denote it by $\tau_V\colon V \to (V^*)^*$. Let V and W be two vector spaces over K. Then, for any linear map $T\colon V \to W$, we have the following naturality condition, represented by a commutative diagram:

$$\begin{tikzcd} V \arrow[r, "\tau_V"] \arrow[d, "T"'] & (V^*)^* \arrow[d, "(T^*)^*"] \\ W \arrow[r, "\tau_W"'] & (W^*)^* \end{tikzcd}$$

Exercise: Let V be a finite-dimensional vector space over K, and let $\mathcal{B} = (v_1, \ldots, v_n)$ be a basis for V. Let $\mathcal{B}^* = (v_1^*, \ldots, v_n^*)$ be the dual basis, and let $(\mathcal{B}^*)^* = ((v_1^*)^*, \ldots, (v_n^*)^*)$ be the basis for $(V^*)^*$ which is dual to $\mathcal{B}^*$. Show that $\tau(\mathcal{B}) = (\mathcal{B}^*)^*$, meaning $\tau(v_i) = (v_i^*)^*$ for all $1 \le i \le n$.

Direct Sums (Section 18.b)

The External Direct Sum

Proposition 18.b.1 (*Vector Space Structure of the Cartesian Product***):** Let V and W be two vector spaces over K. Then, the Cartesian product $V \times W$ becomes a vector space over K, provided we endow it with the following operations:

- $(v_1, w_1) + (v_2, w_2) := (v_1 + v_2, w_1 + w_2)$ for all $(v_1, w_1), (v_2, w_2) \in V \times W$.

- $\alpha \cdot (v, w) := (\alpha v, \alpha w)$ for all $\alpha \in K$ and $(v, w) \in V \times W$.

The zero element of this space is given by $0_{V \times W} := (0_V, 0_W)$. This new vector space $V \times W$ is usually denoted by $V \oplus W$ and is called the direct sum of V and W.

Proof: Exercise. One must verify that all vector space axioms are satisfied under these componentwise operations. ■

Remark (*Canonical Maps associated with Direct Sums*): The direct sum $V \oplus W$ is naturally equipped with several canonical linear maps:

(a) Canonical Embeddings:

- $i_V : V \to V \oplus W$ defined by $i_V(v) := (v, 0)$.
- $i_W : W \to V \oplus W$ defined by $i_W(w) := (0, w)$.

(b) Canonical Projections:

- $p_V : V \oplus W \to V$ defined by $p_V(v, w) := v$.
- $p_W : V \oplus W \to W$ defined by $p_W(v, w) := w$.

Clearly, all these maps are linear. Furthermore, the projections p_V and p_W are surjective, while the embeddings i_V and i_W are injective. By virtue of these embeddings, we can often identify V and W as subspaces of $V \oplus W$ through the identifications $v \leftrightarrow (v, 0)$ and $w \leftrightarrow (0, w)$.

Internal Direct Sums and Complements

Proposition 18.b.2 (*Isomorphism to a Complementary Subspace*): Let V be a vector space over K, and let $U \subseteq V$ be a subspace. Suppose $U' \subseteq V$ is another subspace that acts as a complement of U in V, which implies that $U + U' = V$ and $U \cap U' = \{0\}$. Then, the map

$$\varphi : U \oplus U' \to V, \quad \varphi(u, u') := u + u'$$

is a linear isomorphism.

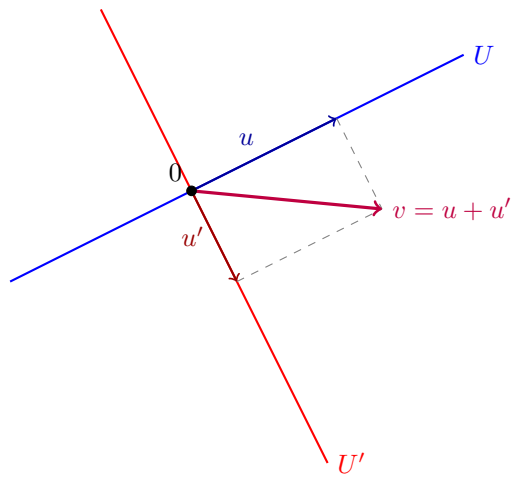
Proof: The linearity of φ follows immediately from the definitions of the operations in $U \oplus U'$ and the linearity of addition in V.

To establish injectivity, suppose $\varphi(u, u') = 0$, which implies $u + u' = 0$, and therefore $u = -u'$. Since $u \in U$ and $u' \in U'$, it follows that $u = -u' \in U \cap U'$. However, as U' is a complement of U, we have $U \cap U' = \{0\}$ by definition. Consequently, $u = u' = 0$, which shows that $\ker(\varphi) = \{0_{U \oplus U'}\}$. Thus, φ is injective.

Regarding surjectivity, we observe that since U' is a complement of U in V, we have $U + U' = V$ by definition. Thus, for any $v \in V$, there exist $u \in U$ and $u' \in U'$ such that $u + u' = v$. This implies $\varphi(u, u') = v$, and therefore $\mathrm{Im}(\varphi) = V$. This proves that φ is surjective, and consequently, an isomorphism. ■

Remark: While $U \oplus U'$ is formally a different vector space from V, the existence of the canonical isomorphism φ allows us to treat V as the direct sum of its internal subspaces whenever a complement is specified.

Geometrically, in $\mathbb{R}^2$, this represents the decomposition of any vector v uniquely into a sum of a vector $u \in U$ and a vector $u' \in U'$, constructed via a parallelogram parallel to the complementary subspaces:



Generalization to Multiple Summands

The concept of a direct sum can be extended to any finite collection of vector spaces $U_1, \ldots, U_m$ over K. We define their direct sum as:

$$U_1 \oplus \cdots \oplus U_m := U_1 \times \cdots \times U_m,$$

endowed with coordinatewise operations.

For an infinite family of vector spaces $\{U_i\}_{i \in I}$, we define the direct sum $\bigoplus_{i \in I} U_i$ as the set of functions $t \colon I \to \bigcup_{i \in I} U_i$ such that $t(i) \in U_i$ for all $i \in I$, and $t(i) \neq 0$ for at most finitely many indices $i \in I$. This ensures that any element of the direct sum can be represented as a finite formal sum of components.

Quotient Spaces (Section 18.c)

Equivalence Relation and the Quotient Space

Let V be a vector space over K, and let $U \subseteq V$ be a subspace. We will define a new vector space over K, denoted by V/U, together with a linear map $\pi \colon V \to V/U$ such that π is surjective and $\ker(\pi) = U$.

Definition (*The Quotient V/U And The Quotient Map π*): The space V/U is called the quotient of V by U, or V modulo U. The map π is called the quotient map, or the projection onto the quotient space.

We begin by defining a relation on V. For $v_1, v_2 \in V$, we define $v_1 \sim v_2$ if and only if $v_1 - v_2 \in U$.

Lemma (*Equivalence Relation on V*): The relation $\sim$ defines an equivalence relation on V.

Proof: Reflexivity: For all $v \in V$, $v \sim v$ because $v - v = 0 \in U$.

Symmetry: If $v_1 \sim v_2$, then $v_1 - v_2 \in U$. Since U is a subspace, $-(v_1 - v_2) = v_2 - v_1 \in U$, which implies $v_2 \sim v_1$.

Transitivity: If $v_1 \sim v_2$ and $v_2 \sim v_3$, then $v_1 - v_2 \in U$ and $v_2 - v_3 \in U$. Because U is a subspace, their sum is in U:

$$(v_1 - v_2) + (v_2 - v_3) = v_1 - v_3 \in U,$$

which means $v_1 \sim v_3$. ■

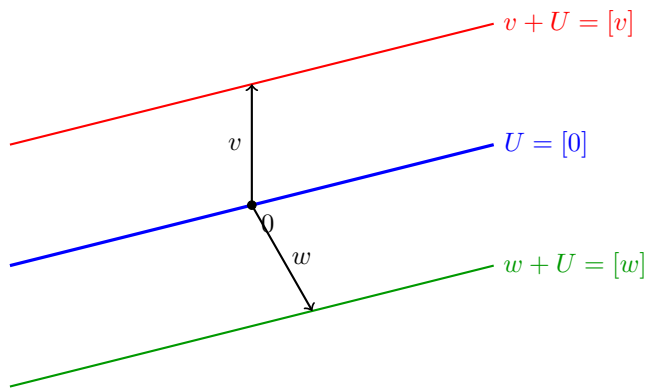
We denote the equivalence class of $v \in V$ by $[v]$. Thus,

$$[v] = v + U = \{v + u \mid u \in U\}.$$

Definition 18.c.1 (*The Quotient Space*): Let V be a vector space over K, and let $U \subseteq V$ be a subspace. We define the quotient space V/U as the set of equivalence classes of the equivalence relation $\sim$.

Alternatively, each element in V/U is a set of the form $v + U$, where $v \in V$.

Geometrically, if $V = \mathbb{R}^2$ and U is a line through the origin, the equivalence classes $[v]$ forming the space V/U are precisely the lines parallel to U:



The Vector Space Structure of V/U

We want to turn V/U into a vector space over K. We define addition as $[v_1] + [v_2] := [v_1 + v_2]$ for all $[v_1], [v_2] \in V/U$. Alternatively, this can be written as $(v_1 + U) + (v_2 + U) := (v_1 + v_2) + U$. For scalar multiplication, let $\alpha \in K$ and $[v] \in V/U$. We define $\alpha \cdot [v] := [\alpha v]$. The zero element is defined as $0_{V/U} := [0_V] = U$.

Proposition 18.c.2 (*Well-definedness of Operations in V/U*): The operations defined above are well-defined. Moreover, they give V/U the structure of a vector space over K.

Proof: To establish well-definedness, we must show that the operations do not depend on the choice of representatives. For scalar multiplication, suppose $[v] = [v']$, and let $\alpha \in K$. We need to show that $[\alpha v] = [\alpha v']$. Indeed, since $[v] = [v']$, we have $v - v' \in U$. Because U is a subspace, $\alpha(v - v') = \alpha v - \alpha v' \in U$. Consequently, $[\alpha v] = [\alpha v']$.

Similarly, for addition, suppose $[v_1] = [v_1']$ and $[v_2] = [v_2']$. We need to show that $[v_1 + v_2] = [v_1' + v_2']$. By assumption, $v_1 - v_1' \in U$ and $v_2 - v_2' \in U$. Adding these yields:

$$(v_1 + v_2) - (v_1' + v_2') = (v_1 - v_1') + (v_2 - v_2') \in U.$$

Therefore, $v_1 + v_2 \sim v_1' + v_2'$, which implies $[v_1 + v_2] = [v_1' + v_2']$. This proves that the operations are well-defined.

It remains to show that they satisfy the axioms of a vector space. These follow directly from the corresponding axioms in V. (Exercise: complete the proof!) ■

We now define a map $\pi \colon V \to V/U$ by $\pi(v) := [v] \in V/U$ for all $v \in V$.

Proposition 18.c.3 (*The Quotient Map*): The map $\pi \colon V \to V/U$ is a linear map. Moreover, it is surjective (i.e., $\mathrm{Im}(\pi) = V/U$), and $\ker(\pi) = U$.

Proof: Linearity: For $v_1, v_2 \in V$, $\pi(v_1 + v_2) = [v_1 + v_2] = [v_1] + [v_2] = \pi(v_1) + \pi(v_2)$. For $\alpha \in K$ and $v \in V$, $\pi(\alpha v) = [\alpha v] = \alpha[v] = \alpha\pi(v)$.

Surjectivity: Let $x \in V/U$. By definition, x is an equivalence class $[v]$ of some element $v \in V$. Thus, $x = \pi(v)$.

Kernel: We claim that $\ker(\pi) = U$. First, let $v \in \ker(\pi)$, meaning $\pi(v) = 0$. Then, $[v] = [0]$, which implies $v = v - 0 \in U$. Hence, $\ker(\pi) \subseteq U$. Conversely, let $u \in U$. Then, $u \sim 0$ because $u - 0 \in U$. Therefore, $\pi(u) = [u] = [0] = 0$, so $u \in \ker(\pi)$. This shows that $U \subseteq \ker(\pi)$, completing the proof. ■

Proposition 18.c.4 (*Dimension of the Quotient Space*): Suppose that V is a finite-dimensional vector space. Then, V/U is also finite-dimensional, and $\dim V/U = \dim V - \dim U$.

Proof: We have seen that $\pi\colon V \to V/U$ is a surjective linear map. This guarantees that V/U is finite-dimensional. (Generally, if $T\colon W' \to W''$ is a surjective linear map and W' is finite-dimensional, then W'' is also finite-dimensional, because the images of a basis of W' span W'').

We now apply the Rank-Nullity Theorem to π:

$$\dim V = \dim \ker(\pi) + \dim \operatorname{Im}(\pi).$$

Since $\ker(\pi) = U$ and $\operatorname{Im}(\pi) = V/U$, we obtain $\dim V/U = \dim V - \dim U$. ■

The Isomorphism Theorem and Universal Property

Theorem 18.c.5 (*The First Isomorphism Theorem*): Let V and W be vector spaces over K, and let $T\colon V \to W$ be a linear map. We define a new map $\overline{T}\colon V/\ker(T) \to \operatorname{Im}(T)$ as follows: For $x \in V/\ker(T)$, choose a representative $v \in V$ such that $x = [v]$. Define $\overline{T}(x) := T(v)$. Then:

(a) $\overline{T}$ is well-defined and is a linear map.

(b) $\overline{T}\colon V/\ker(T) \to \operatorname{Im}(T)$ is an isomorphism.

(c) The following diagram commutes: $T = i \circ \overline{T} \circ \pi$, where $i\colon \operatorname{Im}(T) \to W$ is the inclusion map.

$$\begin{tikzcd}
V \arrow[r, "T"] \arrow[d, "\pi"'] & W \\
V/\ker(T) \arrow[r, "\overline{T}\ (\cong)"] & \operatorname{Im}(T) \arrow[u, "i"']
\end{tikzcd}$$

Proof: **(a)** To show that $\overline{T}$ is well-defined, suppose $x = [v] = [v']$. We must show that $T(v) = T(v')$. Indeed, $[v] = [v']$ implies that $v - v' \in \ker(T)$, meaning $T(v - v') = 0$. By linearity, $T(v) = T(v')$.

The linearity of $\overline{T}$ is straightforward: $\overline{T}(\alpha[v]) = \overline{T}([\alpha v]) = T(\alpha v) = \alpha T(v) = \alpha\overline{T}([v])$. Also, $\overline{T}([v_1] + [v_2]) = \overline{T}([v_1 + v_2]) = T(v_1 + v_2) = T(v_1) + T(v_2) = \overline{T}([v_1]) + \overline{T}([v_2])$.

(b) To show that $\overline{T}$ is injective, let $x \in \ker(\overline{T})$, and write $x = [v]$ for some $v \in V$. Then, $\overline{T}(x) = T(v) = 0$, meaning $v \in \ker(T)$. This implies that $x = [v] = 0_{V/\ker(T)}$.

To show that $\overline{T}$ is surjective, let $w \in \operatorname{Im}(T)$. Then, there exists $v \in V$ such that $T(v) = w$. Consequently, $\overline{T}([v]) = T(v) = w$, proving surjectivity.

(c) The commutativity of the diagram follows immediately from the definitions.

■

Theorem 18.c.6 (*Universal Property of Quotient Spaces*): Let V be a vector space over K, and let $U \subseteq V$ be a subspace. The quotient space V/U has the following universal property: For every vector space W and every linear map $T\colon V \to W$ for which $T(U) = 0$ (i.e., for which $\ker(T) \supseteq U$), there exists a unique linear map $T'\colon V/U \to W$ such that $T = T' \circ \pi$. Moreover, $\ker(T') = \ker(T)/U$.

Remark: In algebraic jargon, every linear map $T\colon V \to W$ that maps U to 0 factors through V/U via $T = T' \circ \pi$, and moreover, in a unique way. We say that T induces the map T', or that T descends to a map $T'\colon V/U \to W$.

Proof: We define $T'\colon V/U \to W$ as follows: let $x \in V/U$ and choose $v \in V$ such that $x = [v]$. Define $T'(x) := T(v)$. The proof that T' is well-defined and linear follows the exact same logic as in the Isomorphism Theorem. The diagram commutes by construction, since $T'(\pi(v)) = T'([v]) = T(v)$.

For uniqueness, suppose $T = T'' \circ \pi$ is another factorization. We will show that $T'' = T'$. Let $x \in V/U$. Since π is surjective, there exists $v \in V$ such that $x = \pi(v)$. We then have $T(v) = T'(\pi(v))$ and $T(v) = T''(\pi(v))$, which immediately yields $T'(x) = T''(x)$. ■

Quotients, Complements, and Dual Spaces

Let V be a vector space over K, and let $U \subseteq V$ be a subspace. Let $W \subseteq V$ be a complement of U (i.e., W is a subspace such that $U \cap W = \{0\}$ and $U + W = V$).

Proposition 18.c.7 (*Canonical Isomorphism from a Complement*): Under the assumptions above, there is a canonical isomorphism $Q\colon W \to V/U$, defined by $Q(w) := [w] \in V/U$ for all $w \in W$.

Remark: The isomorphism Q is canonical only once the complement W is specified. In general, there are many choices of complements to U, and there is no preferred choice.

Proof: Consider the inclusion map $i\colon W \to V$ and the projection map $\pi\colon V \to V/U$. Since $Q = \pi \circ i$, Q is a linear map.

We claim that $\ker(Q) = \{0\}$. Indeed, if $w \in \ker(Q)$, then $[w] = 0$, implying that $w \in U$. However, since $w \in W$, we must have $w \in U \cap W = \{0\}$, and therefore $w = 0$. This shows that $\ker(Q) = \{0\}$.

We also claim that Q is surjective. Let $x \in V/U$ and choose $v \in V$ such that $x = [v]$. Since $V = U + W$, there exist $u \in U$ and $w \in W$ such that $v = w + u$. It follows that $[v] = [w + u] = [w]$, so $Q(w) = x$. This proves that Q is surjective, and consequently, an isomorphism. ■

Quotients, Complements, and Annihilators

Proposition 18.c.8 (*Isomorphism from a Specified Complement*): Let $W \subseteq V$ be a complement of U in V. There exists a canonical isomorphism $Q\colon W \to V/U$ defined by $Q(w) := [w]$.

Proof: Q is the composition of the inclusion $i\colon W \to V$ and the projection $\pi\colon V \to V/U$, and thus is linear. Since $W \cap U = \{0\}$, the kernel of Q is trivial. As $V = U + W$, any $v \in V$ can be written as $w + u$, so $[v] = [w + u] = [w]$, which is $Q(w)$. This proves surjectivity and establishes the isomorphism. ■

Proposition 18.c.9 (*Annihilators and Dual Quotients*): Let $U \subseteq V$ be a subspace, and define the annihilator $U^\perp := \{l \in V^* \mid l|_U = 0\}$. Then:

(a) $U^\perp \subseteq V^*$ is a linear subspace.

(b) There is a canonical isomorphism $(V/U)^* \cong U^\perp$.

(c) If V is finite-dimensional, then $U^* \cong V^*/U^\perp$.

Proof: **(a)** Clearly, the zero functional vanishes on U. For $l_1, l_2 \in U^\perp$ and $\alpha, \beta \in K$, we have $(\alpha l_1 + \beta l_2)(u) = \alpha l_1(u) + \beta l_2(u) = 0$ for all $u \in U$, establishing that $U^\perp$ is a subspace.

(b) We define a map $L\colon (V/U)^* \to U^\perp$ by $L(s) := s \circ \pi$. This map is well-defined because for any $u \in U$, $(s \circ \pi)(u) = s([u]) = s(0) = 0$. To show linearity, let $s, t \in (V/U)^*$ and $\alpha, \beta \in K$; then

$$L(\alpha s + \beta t) = (\alpha s + \beta t) \circ \pi = \alpha(s \circ \pi) + \beta(t \circ \pi) = \alpha L(s) + \beta L(t).$$

We define an inverse $P\colon U^\perp \to (V/U)^*$ such that for $T \in U^\perp$, $P(T) = T'$, where T' is the unique map induced by the Universal Property of Quotients such that $T = T' \circ \pi$. The identities $L \circ P = \mathrm{id}$ and $P \circ L = \mathrm{id}$ confirm L is a canonical isomorphism.

(c) Assume V is finite-dimensional. Define the restriction map $R\colon V^* \to U^*$ by $R(\varphi) := \varphi|_U$. This map is linear, as $R(\alpha\varphi + \beta\psi) = (\alpha\varphi + \beta\psi)|_U = \alpha R(\varphi) + \beta R(\psi)$. By extending a basis of U to V, we can construct a functional on V that matches any functional on U, showing R is surjective. Since $\ker(R) = \{\varphi \in V^* \mid \varphi|_U = 0\} = U^\perp$, the First Isomorphism Theorem yields $V^*/U^\perp \cong U^*$.

■

Exercise: Let $T\colon V \to W$ be a linear map between vector spaces. Prove that the annihilator of the image of T is equal to the kernel of its dual map:

$$(\operatorname{Im} T)^\perp = \ker(T^*).$$

Miscellaneous Topics and Motivation (Chapter 20)

Geometric Motivation: Why Abstraction Matters (Section 20.a)

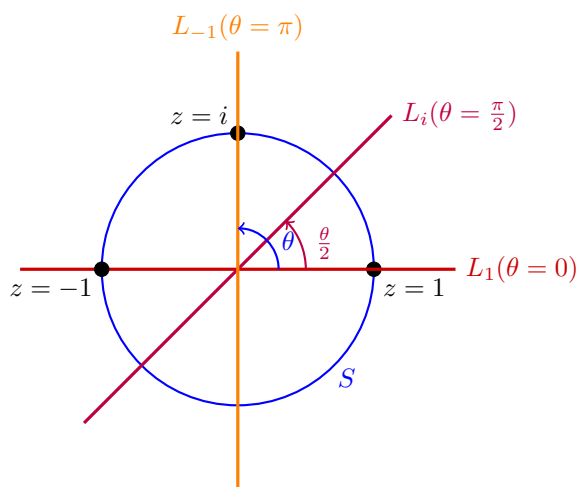
It is a common temptation to simply identify every n-dimensional vector space over K with the standard coordinate space K^n. However, while they are algebraically isomorphic for any fixed parameter, they may not be "continuously" isomorphic when the vector spaces depend dynamically on parameters.

The Möbius Vector Bundle

Definition 20.a.1 (*A Continuous Family of Subspaces*): Let $S \subseteq \mathbb{C}$ be the unit circle defined by $S = \{e^{i\theta} \mid \theta \in [0, 2\pi]\}$. For each $z = e^{i\theta} \in S$, we define a 1-dimensional subspace $L_z \subseteq \mathbb{R}^2$ as follows:

$$L_z := \mathrm{Sp}\left(\cos\frac{\theta}{2}, \sin\frac{\theta}{2}\right).$$

Notice a topological anomaly: $z = 1$ corresponds to both $\theta = 0$ and $\theta = 2\pi$. In the first case ($\theta = 0$), the spanning vector is $(1, 0)$, so $L_1 = \mathrm{Sp}(1, 0)$. In the second case ($\theta = 2\pi$), the spanning vector is $(-1, 0)$, which gives the exact same subspace $L_1 = \mathrm{Sp}(-1, 0)$. The line L_z rotates at exactly half the speed of the point z.



Proposition 20.a.2 (*Non-triviality of the Möbius Bundle*): There does not exist a family of linear isomorphisms $T_z \colon L_z \to \mathbb{R}$ that depends continuously on the parameter $z \in S$.

Proof Sketch: Consider the disjoint union of all these lines parameterized by z, which forms a space $M = \{(z, v) \mid z \in S, v \in L_z\}$. Topologically, this forms a vector bundle over S^1 known as the Möbius strip.

If such a continuous family of isomorphisms T_z existed, it would allow us to construct a global homeomorphism $\Phi \colon M \to S \times \mathbb{R}$. Removing the zero-section (the center circle where $v = 0$) from both spaces would imply that $M \setminus \{0\} \cong S \times (\mathbb{R} \setminus \{0\})$.

However, $S \times (\mathbb{R} \setminus \{0\})$ consists of two disconnected cylinders (an "upper" and "lower" half). In contrast, if you cut a Möbius strip down its center line, it remains a single connected piece. This topological contradiction demonstrates that we cannot consistently and continuously identify all L_z with $\mathbb{R}$. Abstraction is therefore necessary to handle "twisted" families of spaces. ■

Affine Geometry (Section 20.b)

In our strict definition of a vector space, there is always a distinguished origin 0. Affine geometry studies spaces that "look like" vector spaces locally, but have forgotten their origin.

Definition 20.b.1 (*Affine Space*): A subset $X \subseteq V$ is called an **affine space** if there exists a linear subspace $U \subseteq V$ and a translation vector $x_0 \in V$ such that $X = x_0 + U$.

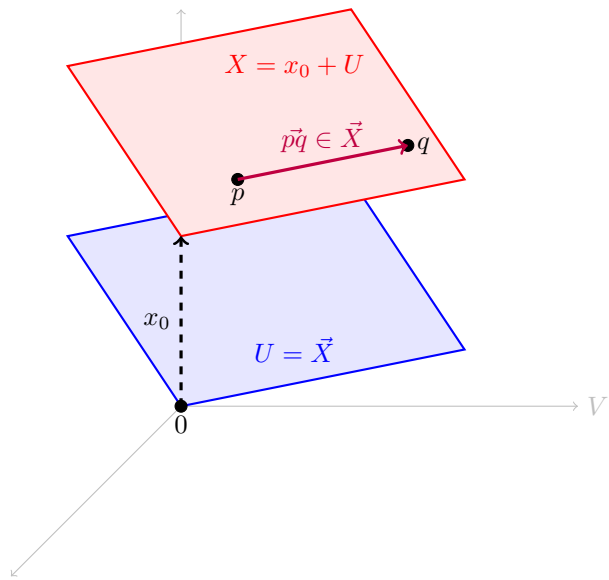
We say that X is modeled on the subspace U, and we write the direction space as $\vec{X} := U$. The elements of X are called *points*, while the elements of $\vec{X}$ are called *vectors*.

Proposition 20.b.2 (*Algebra of Affine Spaces*)**:** Let X be an affine space modeled on $\vec{X}$. Then:

(a) We cannot inherently add two points $p, q \in X$, as $p + q \notin X$ in general.

(b) We can translate a point $p \in X$ by a vector $v \in \vec{X}$ to obtain a new point $p + v \in X$.

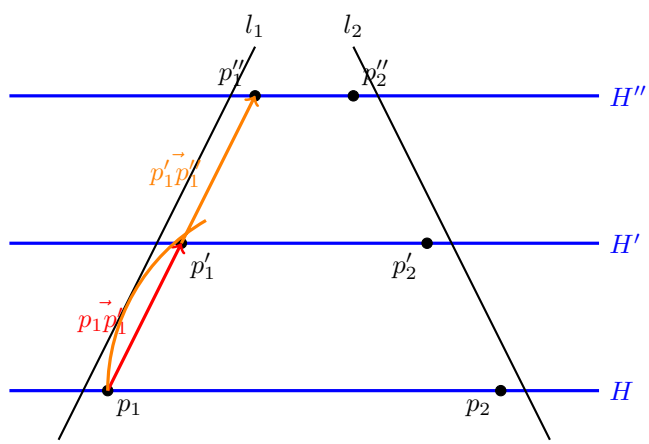
(c) For any two points $p, q \in X$, their difference defines a unique vector $\vec{pq} := q - p \in \vec{X}$.



Theorem 20.b.3 (*Thales' Theorem in Affine Geometry*)**:** Let H, H', H'' be three distinct parallel affine planes in an ambient space V. Let l_1, l_2 be two affine lines intersecting these planes at the points p_1, p_1', p_1'' and p_2, p_2', p_2'' respectively. Then:

$$\frac{\vec{p_1 p_1''}}{\vec{p_1 p_1'}} = \frac{\vec{p_2 p_2''}}{\vec{p_2 p_2'}}.$$

Crucially, the ratio of these vectors does not depend on the specific transversals chosen, but only on the relative distances between the parallel planes.



Introduction to Eigenvalues: Fibonacci Sequences (Section 20.c)

The Vector Space of Fibonacci Sequences

Consider the set $Fib \subseteq \mathbb{R}^\infty$ of all infinite real sequences $(a_0, a_1, a_2, \dots)$ that satisfy the linear recurrence relation $a_n = a_{n-1} + a_{n-2}$ for all $n \geq 2$.

Proposition 20.c.1 (*Structure of Fib*)**:** The set Fib is a 2-dimensional subspace of $\mathbb{R}^\infty$. The evaluation map $F\colon Fib \to \mathbb{R}^2$ sending a sequence to its first two terms, $(a_n) \mapsto (a_0, a_1)$, is a linear isomorphism.

To find a closed-form formula for the Fibonacci numbers (like Binet's formula), we study the **shift map** $S\colon \mathbb{R}^\infty \to \mathbb{R}^\infty$ defined by shifting the sequence one position to the left:

$$S(a_0, a_1, a_2, \dots) := (a_1, a_2, a_3, \dots).$$

Since the shifted sequence also satisfies the Fibonacci recurrence, this map restricts to a linear endomorphism $S'\colon Fib \to Fib$.

By utilizing the isomorphism F, we can translate the abstract shift operator S' into concrete matrix multiplication on $\mathbb{R}^2$. The sequence (a_0, a_1) shifts to $(a_1, a_2) = (a_1, a_0 + a_1)$, which corresponds to multiplication by the matrix $A = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$.

$$\begin{tikzcd}
Fib \arrow[r, "S'"] \arrow[d, "F"', "\cong"] & Fib \arrow[d, "F", "\cong"'] \\
\mathbb{R}^2 \arrow[r, "T_A"'] & \mathbb{R}^2
\end{tikzcd}$$

Eigenvalues and Eigenvectors

The problem of finding a geometric sequence $(1, \lambda, \lambda^2, \dots)$ inside Fib is equivalent to finding sequences that only scale by λ when shifted. That is, we seek vectors v such that $S'(v) = \lambda v$.

Definition 20.c.2 (*Eigenvalues and Eigenvectors*)**:** Let V be a vector space and $T \in \mathrm{End}(V)$. A scalar $\lambda \in K$ is called an **eigenvalue** of T if there exists a non-zero vector $v \in V$ such that:

$$T(v) = \lambda v.$$

The vector v is called an **eigenvector** associated with the eigenvalue λ.

In the context of the Fibonacci matrix $A = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, calculating the roots of the characteristic polynomial reveals the eigenvalues to be the Golden Ratio $\varphi = \frac{1+\sqrt{5}}{2}$ and its conjugate $\psi = \frac{1-\sqrt{5}}{2}$. Their associated eigenvectors span Fib, leading directly to the closed-form Binet formula for the sequence. This forms the primary motivation for the upcoming deeper study of eigenvalues and determinants.

Determinants (Chapter 20)

Introduction and Motivation (Section 20.a)

In linear algebra, determining whether a matrix is invertible is a fundamental problem. Let us begin by analyzing the familiar 2×2 case. Let $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in M_{2\times 2}(K)$. There is a very simple, well-known algebraic criterion to check whether A is invertible.

Proposition 20.a.1 (*Invertibility of* 2×2 *Matrices*): The matrix A is invertible if and only if $ad - bc \neq 0$. Moreover, if A is invertible, then its inverse is given by the explicit formula:

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

The scalar quantity $ad - bc$ is called the determinant of A, denoted $\det(A)$.

Proof: First, we show that A is not invertible if and only if its rank is strictly less than 2. Because A is a 2×2 matrix, having a rank less than 2 implies that its column vectors are linearly dependent.

Remark (*Exercise*): Check that the columns being linearly dependent implies that either the first column is the zero vector (i.e., $\binom{a}{c} = \binom{0}{0}$) or the second column is a scalar multiple of the first (i.e., $\binom{b}{d} = \tau\binom{a}{c}$ for some scalar $\tau \in K$).

If $\binom{a}{c} = \lambda\binom{b}{d}$, then $a = \lambda b$ and $c = \lambda d$. Consequently, $ad - bc = (\lambda b)d - b(\lambda d) = 0$. If, on the other hand, $\binom{b}{d} = \tau\binom{a}{c}$, then $b = \tau a$ and $d = \tau c$. Consequently, $ad - bc = a(\tau c) - (\tau a)c = 0$. Thus, we have proved that if A is not invertible, then $ad - bc = 0$.

Conversely, assume $ad - bc = 0$. If $a = 0$, then $bc = 0$, so either $b = 0$ or $c = 0$. This means either $A = \left(\begin{smallmatrix} 0 & 0 \\ c & d \end{smallmatrix}\right)$ or $A = \left(\begin{smallmatrix} 0 & b \\ 0 & d \end{smallmatrix}\right)$. In both of these cases, A clearly has linearly dependent columns and is therefore not invertible.

Now assume $a \neq 0$. This implies $d = \frac{bc}{a}$, and thus we can write $A = \left(\begin{smallmatrix} a & b \\ c & \frac{bc}{a} \end{smallmatrix}\right)$. We can observe that the second column of A is exactly a multiple of the first column because $\left(\begin{smallmatrix} b \\ \frac{bc}{a} \end{smallmatrix}\right) = \frac{b}{a}\binom{a}{c}$. This implies that the columns of A are linearly dependent. Hence, $\operatorname{rank}(A) < 2$, which implies A is not invertible. ■

To confirm the sufficiency of this condition and to find the explicit inverse, the formula for A^{-1} can be verified by direct matrix multiplication:

$$\frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \frac{1}{ad - bc} \begin{pmatrix} ad - bc & bd - bd \\ -ac + ac & -bc + ad \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Goals: While the 2×2 formula is easy to memorize, it does not immediately suggest how to handle larger matrices. Our main goals now are to make the determinant conceptually rigorous and to generalize the determinant function to arbitrary $n \times n$ matrices.

Multilinear and Alternating Functions

To generalize the determinant, we should view an $n \times n$ matrix not just as a grid of numbers, but as a collection of n row vectors.

Definition 20.a.2 (*n-Linearity*): Let $D : M_{n\times n}(K) \to K$ be a function. We say that D is *n-linear* if for every $1 \leq i \leq n$, D is linear as a function of the i-th row when all the other $n - 1$ rows are held fixed.

We can think of the matrix space $M_{n\times n}(K)$ as a Cartesian product of row vectors: $K^n_{\text{row}} \times \cdots \times K^n_{\text{row}}$. If A has rows $\alpha_1, \ldots, \alpha_n$, we write $D(A) = D(\alpha_1, \ldots, \alpha_n)$. Saying that D is n-linear means that for any chosen row index i, the following two linearity conditions hold:

(a) $D(\alpha_1, \ldots, \alpha_i + \alpha_i', \ldots, \alpha_n) = D(\alpha_1, \ldots, \alpha_i, \ldots, \alpha_n) + D(\alpha_1, \ldots, \alpha_i', \ldots, \alpha_n)$.

(b) $D(\alpha_1, \ldots, c \cdot \alpha_i, \ldots, \alpha_n) = c \cdot D(\alpha_1, \ldots, \alpha_i, \ldots, \alpha_n)$ for all scalars $c \in K$.

Example: For a matrix $A \in M_{n\times n}(K)$, write $A(i,j)$ for its entry at place i,j. Let $k_1,\dots,k_n$ be a fixed sequence of integers such that $1 \le k_i \le n$ for all i. Fix also a scalar $c \in K$. Consider $D : M_{n\times n}(K) \to K$ defined by the product of one entry from each row:

$$D(A) := c \cdot A(1,k_1) \cdot A(2,k_2) \cdots A(n,k_n).$$

We claim that D is an n-linear function.

To see this, consider $D(\alpha_1,\dots,\alpha_i,\dots,\alpha_n) = b \cdot A(i,k_i)$, where $b \in K$ is the product of all the fixed terms. Notice that b depends only on the rows $\alpha_1,\dots,\alpha_{i-1},\alpha_{i+1},\dots,\alpha_n$ and does not depend on row i. Let $\alpha_i' = (\alpha_{i,1}',\dots,\alpha_{i,n}')$ be another possible i-th row. Then, by standard distribution:

$$D(\dots,\alpha_i+\alpha_i',\dots) = b \cdot (A(i,k_i) + \alpha_{i,k_i}') = D(\dots,\alpha_i,\dots) + D(\dots,\alpha_i',\dots).$$

Similarly, for any scalar $\lambda \in K$:

$$D(\dots,\lambda\cdot\alpha_i,\dots) = b \cdot (\lambda A(i,k_i)) = \lambda \cdot D(\dots,\alpha_i,\dots).$$

Thus, D is indeed n-linear.

Example (*Diagonal Product*): A more concrete example: $D(A) := A_{11} \cdot A_{22} \cdot \dots \cdot A_{nn}$ (i.e., the product of the terms on the main diagonal) is an n-linear function.

Now let us try to find all 2-linear functions $D : M_{2\times 2}(K) \to K$. Write the identity matrix as $I = \left(\begin{smallmatrix} -\epsilon_1- \\ -\epsilon_2- \end{smallmatrix}\right)$, where $\epsilon_1 = (1,0)$ and $\epsilon_2 = (0,1)$.

If D is 2-linear, then we have:

$$\begin{aligned} D(A) &= D(A_{11}\epsilon_1 + A_{12}\epsilon_2, A_{21}\epsilon_1 + A_{22}\epsilon_2) \\ &= A_{11}D(\epsilon_1, A_{21}\epsilon_1 + A_{22}\epsilon_2) + A_{12}D(\epsilon_2, A_{21}\epsilon_1 + A_{22}\epsilon_2) \\ &= A_{11}A_{21}D(\epsilon_1,\epsilon_1) + A_{11}A_{22}D(\epsilon_1,\epsilon_2) + A_{12}A_{21}D(\epsilon_2,\epsilon_1) + A_{12}A_{22}D(\epsilon_2,\epsilon_2). \end{aligned}$$

So D is completely determined by its values on the four pairings: $D(\epsilon_1,\epsilon_1)$, $D(\epsilon_1,\epsilon_2)$, $D(\epsilon_2,\epsilon_1)$, and $D(\epsilon_2,\epsilon_2)$.

Lemma 20.a.3 (*Closure of n-Linear Functions*): Any linear combination of n-linear functions is again an n-linear function.

Proof: It is enough to prove this for a linear combination of two n-linear functions. Let $D, E : M_{n\times n}(K) \to K$ be n-linear functions and $a, b \in K$. We define the combined function as $(aD + bE)(A) := aD(A) + bE(A)$. By expanding $(aD+bE)(\dots,\alpha_i+\alpha_i',\dots)$ and $(aD+bE)(\dots,\lambda\alpha_i,\dots)$ using the individual linearities of D and E, we see the result naturally holds. ■

Example: Consider $D : M_{2\times 2}(K) \to K$ defined by $D(A) := A_{11}A_{22} - A_{12}A_{21}$. The function D is 2-linear because it can be written as $D = D_1 + D_2$, where $D_1(A) = A_{11}A_{22}$ and $D_2(A) = -A_{12}A_{21}$, both of which are 2-linear by our previous example. Furthermore, $D(I) = 1$, and if A' is obtained from A by interchanging the rows, then $D(A') = -D(A)$.

Indeed, $D(A') = D\left(\begin{smallmatrix} A_{21} & A_{22} \\ A_{11} & A_{12} \end{smallmatrix}\right) = A_{21} \cdot A_{12} - A_{22}A_{11} = -(A_{11}A_{22} - A_{12}A_{21}) = -D(A)$.

Definition 20.a.4 (*Alternating Property*): Let $D : M_{n\times n}(K) \to K$ be an n-linear function. We say that D is *alternating* if for every matrix A in which some two rows are equal, we have $D(A) = 0$.

Proposition 20.a.5 (*Adjacent Swap Property*): Let D be an n-linear function. Assume that D has the property that whenever A has two equal adjacent rows, then $D(A) = 0$. Then the following are equivalent:

(a) D is alternating.

(b) For any matrix B, if B' is obtained from B by interchanging two of its rows, then $D(B') = -D(B)$.

Proof: We begin with the proof of (2). Assume first that B' is obtained from B by interchanging two adjacent rows (say, row k and row $k+1$). Consider evaluating D on a matrix where both the k-th and $(k+1)$-th rows are equal to $\beta_k + \beta_{k+1}$. By our assumption, this yields 0. Expanding this via n-linearity gives:

$$D(\dots, \beta_k + \beta_{k+1}, \beta_k + \beta_{k+1}, \dots) = D(\dots, \beta_k, \beta_k, \dots) + D(\dots, \beta_k, \beta_{k+1}, \dots)$$
$$+D(\dots, \beta_{k+1}, \beta_k, \dots) + D(\dots, \beta_{k+1}, \beta_{k+1}, \dots) = 0.$$

By assumption, the terms with adjacent equal rows are zero, leaving $D(B) + D(B') = 0$, which naturally implies $D(B') = -D(B)$.

Now suppose B' is obtained from B by interchanging rows k and l where $k < l$. We can obtain B' by doing a sequence of adjacent row interchanges. Moving row l to position k requires $r = l - k$ adjacent interchanges. Next, moving the original row k down to position l requires $r - 1$ adjacent interchanges. The total number of interchanges is $2r - 1$. Since this is always an odd number, we flip the sign an odd number of times. Thus, $D(B') = (-1)^{2r-1}D(B) = -D(B)$.

We now prove (1). Let A be a matrix in which row i equals row j where $i < j$. If $j = i + 1$, then by assumption $D(A) = 0$. If $j > i + 1$, we interchange row j with row $i + 1$ to obtain a matrix A' with two equal adjacent rows. By assumption $D(A') = 0$, but by (2), $D(A') = -D(A)$, which implies $D(A) = 0$. ■

The Determinant Function

Definition 20.a.6 (*Determinant Function*): A function $D : M_{n\times n}(K) \to K$ is called a *determinant function* if it is n-linear, it is alternating, and it normalizes the identity matrix such that $D(I) = 1$.

Example (*Warmup*): Let us find all determinant functions on $M_{2\times 2}(K)$. Write $I = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} -\epsilon_1- \\ -\epsilon_2- \end{smallmatrix}\right)$, meaning $\epsilon_1 = (1, 0)$ and $\epsilon_2 = (0, 1)$. If D is a 2-linear function, we can expand it algebraically:

$$\begin{aligned} D(A) &= D(A_{11}\epsilon_1 + A_{12}\epsilon_2, A_{21}\epsilon_1 + A_{22}\epsilon_2) \\ &= A_{11}D(\epsilon_1, A_{21}\epsilon_1 + A_{22}\epsilon_2) + A_{12}D(\epsilon_2, A_{21}\epsilon_1 + A_{22}\epsilon_2) \\ &= A_{11}A_{21}D(\epsilon_1, \epsilon_1) + A_{11}A_{22}D(\epsilon_1, \epsilon_2) \\ &\quad + A_{12}A_{21}D(\epsilon_2, \epsilon_1) + A_{12}A_{22}D(\epsilon_2, \epsilon_2). \end{aligned}$$

So, D is completely determined by its values on these standard basis row pairings.

If D is alternating, then evaluating equal rows gives zero, meaning $D(\epsilon_1, \epsilon_1) = D(\epsilon_2, \epsilon_2) = 0$. Additionally, swapping rows flips the sign, so $D(\epsilon_2, \epsilon_1) = -D(\epsilon_1, \epsilon_2) = -D(I)$. Finally, if D is a determinant function, $D(I) = 1$. Substituting these values back into our expansion, we perfectly recover the familiar formula:

$$D(A) = A_{11}A_{22}(1) + A_{12}A_{21}(-1) = A_{11}A_{22} - A_{12}A_{21}.$$

Therefore, the 2×2 determinant function is completely unique!

Definition 20.a.7 (*Submatrix*): Let $A \in M_{n\times n}(K)$, $n \geq 2$, and $1 \leq i \leq n$, $1 \leq j \leq n$. We denote by $A(i|j) \in M_{(n-1)\times(n-1)}(K)$ the matrix obtained from A by deleting row i and column j. If D is an $(n-1)$-linear function, we intuitively define $D_{ij}(A) := D(A(i|j))$.

Theorem 20.a.8 (*Recursive Construction of Determinants*): Let $n \geq 2$, and let D be an $(n-1)$-linear function which is alternating. Let $1 \leq j \leq n$. Define $E_j : M_{n\times n}(K) \to K$ by

$$E_j(A) := \sum_{i=1}^{n} (-1)^{i+j} A_{ij} D_{ij}(A).$$

Then, E_j is an n-linear function and it is alternating. Furthermore, if D is a determinant function, then so is E_j.

Proof: Clearly, $D_{ij}(A)$ is independent of the i-th row of A because that row is deleted in the construction of $A(i|j)$. Since D is $(n-1)$-linear, the map $A \mapsto D_{ij}(A)$ is linear when viewed as a function of each of its remaining rows. The function $A \mapsto A_{ij}D_{ij}(A)$ is therefore n-linear. Since linear combinations of n-linear functions are n-linear, $E_j(A)$ is n-linear too.

We show that E_j is alternating. By Proposition 20.a.5, it is enough to show $E_j(A) = 0$ whenever A has two equal adjacent rows. Assume $\alpha_k = \alpha_{k+1}$ for some index k. For any $i \neq k, k+1$, the submatrix $A(i|j)$ still has two equal rows, hence $D_{ij}(A) = 0$. This leaves only two non-zero terms in the sum:

$$E_j(A) = (-1)^{k+j} A_{kj} D_{kj}(A) + (-1)^{k+1+j} A_{(k+1)j} D_{(k+1)j}(A).$$

But, since $\alpha_k = \alpha_{k+1}$, we have $A_{kj} = A_{(k+1)j}$ and $A(k|j) = A(k+1|j)$, meaning $D_{kj}(A) = D_{(k+1)j}(A)$. Thus, the two terms perfectly cancel out, yielding $E_j(A) = 0$. This proves E_j is alternating.

Finally, assume D is a determinant function, meaning $D(I_{n-1}) = 1$. Note that $I_n(j|j) = I_{n-1}$ for all j. When computing $E_j(I_n)$, the term $(I_n)_{ij}$ is 1 if $i = j$, and 0 if $i \neq j$. Therefore, the sum collapses to a single term:

$$E_j(I_n) = (-1)^{j+j} D_{jj}(I_n) = D(I_n(j|j)) = 1.$$

This shows that E_j is a valid determinant function. ■

Corollary 20.a.9 (*Existence of Determinants*): For all $n \geq 1$, at least one determinant function exists.

Example: Consider $A \in M_{3\times 3}(K)$. Using the uniquely defined 2×2 determinant function $|B| := B_{11}B_{22} - B_{12}B_{21}$, we can explicitly write out $E_1(A)$ as:

$$E_1(A) = A_{11} \begin{vmatrix} A_{22} & A_{23} \\ A_{32} & A_{33} \end{vmatrix} - A_{21} \begin{vmatrix} A_{12} & A_{13} \\ A_{32} & A_{33} \end{vmatrix} + A_{31} \begin{vmatrix} A_{12} & A_{13} \\ A_{22} & A_{23} \end{vmatrix}.$$

We can similarly construct $E_2(A)$ and $E_3(A)$ along the other columns, and all of these form valid determinant functions on $M_{3\times 3}(K)$.

Uniqueness of the Determinant (Section 20.b)

We have shown that a determinant function exists. Now, we will prove that it is absolutely unique. Let $D : M_{n\times n}(K) \to K$ be an n-linear alternating function. Let $A \in M_{n\times n}(K)$ be a matrix with rows $\alpha_1, \ldots, \alpha_n$. We denote the identity matrix as I, whose rows are the standard basis vectors $\epsilon_1, \ldots, \epsilon_n$ of K^n_{row}.

Clearly, we can express each row of A in terms of the standard basis: $\alpha_i = \sum_{j=1}^n A(i,j)\epsilon_j$ for $1 \leq i \leq n$. Using the n-linearity of D, we can expand $D(A)$ as follows:

$$\begin{aligned} D(A) &= D(\alpha_1, \ldots, \alpha_n) \\ &= D\left(\sum_{j=1}^n A(1,j)\epsilon_j, \alpha_2, \ldots, \alpha_n\right) \\ &= \sum_{j=1}^n A(1,j) \cdot D(\epsilon_j, \alpha_2, \ldots, \alpha_n) \\ &= \sum_{j=1}^n A(1,j) \cdot D\left(\epsilon_j, \sum_{k=1}^n A(2,k)\epsilon_k, \ldots, \alpha_n\right) \\ &= \sum_{j,k} A(1,j) \cdot A(2,k) \cdot D(\epsilon_j, \epsilon_k, \ldots, \alpha_n) \end{aligned}$$

where both j and k run between 1 and n. Continuing this expansion for all n rows yields an enormous sum:

$$D(A) = \sum_{k_1,\ldots,k_n} A(1,k_1) \cdot A(2,k_2) \cdots A(n,k_n) \cdot D(\epsilon_{k_1}, \epsilon_{k_2}, \ldots, \epsilon_{k_n})$$

where each of the indices $k_1, \ldots, k_n$ runs between 1 and n.

Thus far, we have not used the fact that D is alternating. Because D is alternating, the term $D(\epsilon_{k_1}, \ldots, \epsilon_{k_n}) = 0$ whenever two of the indices $k_1, \ldots, k_n$ are equal. Thus, we are interested only in ordered sequences $(k_1, \ldots, k_n)$ in which $k_i \in \{1, \ldots, n\}$ for all i, and $k_i \neq k_j$ for all $i \neq j$.

Such a sequence is called a *permutation* of $\{1, \ldots, n\}$, or a permutation of degree n. Alternatively, we can think of a permutation of degree n as a bijective function $\sigma : \{1, \ldots, n\} \to \{1, \ldots, n\}$. Then, the sequence $(k_1, \ldots, k_n) = (\sigma(1), \sigma(2), \ldots, \sigma(n))$ is a permutation as defined earlier. In other words, a permutation of degree n is simply a re-ordering of a list of n elements: $(1, \ldots, n) \mapsto (\sigma(1), \ldots, \sigma(n))$.

We can nicely rewrite our expansion of $D(A)$ as:

$$D(A) = \sum_{\sigma} A(1, \sigma(1)) \cdot A(2, \sigma(2)) \cdots A(n, \sigma(n)) \cdot D(\epsilon_{\sigma(1)}, \epsilon_{\sigma(2)}, \ldots, \epsilon_{\sigma(n)})$$

where σ runs over all possible permutations of degree n.

Properties of Permutations

An important property about permutations is that if σ is a permutation of degree n, one can pass from the sequence $(1, \ldots, n)$ to $(\sigma(1), \ldots, \sigma(n))$ by performing a finite sequence of interchanges of pairs.

More precisely, a *transposition* is a permutation $\theta : \{1, \ldots, n\} \to \{1, \ldots, n\}$ for which there exist $i, j \in \{1, \ldots, n\}$ with $i \neq j$ such that $\theta(i) = j$, $\theta(j) = i$, and $\theta(k) = k$ for all $k \neq i, j$.

Every permutation σ can be written as a composition of transpositions:

$$\sigma = \theta_1 \circ \cdots \circ \theta_m \tag{20.1}$$

where $\theta_1, \ldots, \theta_m$ are transpositions. However, these θ_i's and the total number m are not unique. The same permutation σ can be written as $\sigma = \theta_1 \circ \cdots \circ \theta_m = \theta'_1 \circ \cdots \circ \theta'_{m'}$.

Remark: Why is the decomposition (20.1) always possible? Start with the sorted list $(1, \ldots, n)$. If $\sigma(1) \neq 1$, then interchange 1 and $\sigma(1)$. We now have $(\sigma(1), \ldots, 1, \ldots, n)$. Next, look at the second position. If it is not $\sigma(2)$, interchange it with $\sigma(2)$, and proceed similarly down the line. Of course, there are other ways to do it. For example, the permutation $(3, 2, 1)$ can be obtained in 3 steps:

$$(1, 2, 3) \to (2, 1, 3) \to (2, 3, 1) \to (3, 2, 1)$$

But, it can also be obtained in just one step: $(1, 2, 3) \to (3, 2, 1)$.

Lemma 20.b.1 (*Parity of Transpositions*): If $(\sigma(1), \ldots, \sigma(n))$ is obtained from $(1, \ldots, n)$ in two different ways—one time by a sequence of m transpositions, and another time by a sequence of m' transpositions—then m and m' must have the same parity. That is, m' is even if and only if m is even, and m' is odd if and only if m is odd.

In other words, $(-1)^{m'} = (-1)^m$. The parity depends only on the permutation σ, and not on the particular sequence of transpositions used.

A permutation σ is called *even* if m is even, and *odd* if m is odd. The number $\operatorname{sgn}(\sigma) := (-1)^m \in \{-1, 1\}$ is called the *sign* of σ.

Proof of Lemma: We know from previous results that determinant functions $E : M_{n \times n}(K) \to K$ exist for $n \geq 1$. Let us fix one such determinant function E (we denote it by E to distinguish it from D in the previous discussion).

Consider the evaluation $E(\epsilon_{\sigma(1)}, \ldots, \epsilon_{\sigma(n)})$. If $(\sigma(1), \ldots, \sigma(n))$ can be obtained from $(1, \ldots, n)$ by a sequence of m transpositions, then the matrix with rows $\epsilon_{\sigma(1)}, \ldots, \epsilon_{\sigma(n)}$ can be obtained from the identity matrix I by a sequence of m interchanges of pairs of rows.

Because E is alternating, swapping two rows flips the sign of the output. Therefore:

$$E(\epsilon_{\sigma(1)}, \ldots, \epsilon_{\sigma(n)}) = (-1)^m E(\epsilon_1, \ldots, \epsilon_n) = (-1)^m E(I) = (-1)^m.$$

But, this exact same logic applies to the sequence of m' transpositions. Thus:

$$E(\epsilon_{\sigma(1)}, \ldots, \epsilon_{\sigma(n)}) = (-1)^{m'} E(\epsilon_1, \ldots, \epsilon_n) = (-1)^{m'} E(I) = (-1)^{m'}.$$

This identically implies that $(-1)^m = (-1)^{m'}$, proving the lemma. ■

Returning to our original function $D(A)$, we have seen that:

$$D(A) = \sum_{\sigma} A(1, \sigma(1)) \cdots A(n, \sigma(n)) \cdot D(\epsilon_{\sigma(1)}, \ldots, \epsilon_{\sigma(n)}).$$

By the alternating property of D, performing the transpositions necessary to sort $(\sigma(1), \ldots, \sigma(n))$ back to $(1, \ldots, n)$ yields:

$$D(\epsilon_{\sigma(1)}, \ldots, \epsilon_{\sigma(n)}) = \operatorname{sgn}(\sigma) \cdot D(\epsilon_1, \ldots, \epsilon_n) = \operatorname{sgn}(\sigma) \cdot D(I). \tag{20.2}$$

Theorem 20.b.2 (*Leibniz Formula & Uniqueness*): For all $n \geq 1$, there exists a unique determinant function on $M_{n\times n}(K)$. We will denote it by det. It is mathematically given by the formula:

$$\det(A) = \sum_{\sigma} \operatorname{sgn}(\sigma) \cdot A(1, \sigma(1)) \cdot A(2, \sigma(2)) \cdots A(n, \sigma(n)).$$

Proof: This follows immediately from (20.2). Indeed, if D is a determinant function, then $D(I) = 1$. Substituting this into (20.2) yields the exact formula for $D(A)$ stated in the theorem, proving that the function is entirely unique. ■

Corollary 20.b.3 (*Scaling of Alternating Functions*): For any n-linear alternating function $D : M_{n\times n}(K) \to K$, we have $D(A) = \det(A) \cdot D(I)$ for all $A \in M_{n\times n}(K)$.

The Symmetric Group

Denote by S_n the set of permutations $\sigma : \{1, \ldots, n\} \to \{1, \ldots, n\}$. The set S_n has the algebraic structure of a group under composition: if $\sigma, \tau \in S_n$, their composition is $(\sigma \circ \tau)(x) = \sigma(\tau(x))$. The identity function id is the neutral element. The inverse σ^{-1} exists because σ is bijective. Furthermore, the size of the symmetric group is $|S_n| = n!$ (Exercise).

Lemma 20.b.4 (*Multiplicativity of Signum*): For all $\sigma, \tau \in S_n$, we have $\operatorname{sgn}(\sigma\circ\tau) = \operatorname{sgn}(\sigma)\cdot\operatorname{sgn}(\tau)$.

Proof: Suppose $\sigma = \theta_1 \circ \cdots \circ \theta_m$ and $\tau = \lambda_1 \circ \cdots \circ \lambda_l$, where the θ_i's and λ_j's are all transpositions. Then, their composition is simply $\sigma \circ \tau = \theta_1 \circ \cdots \circ \theta_m \circ \lambda_1 \circ \cdots \circ \lambda_l$, which explicitly consists of $m + l$ transpositions. Therefore:

$$\operatorname{sgn}(\sigma \circ \tau) = (-1)^{m+l} = (-1)^m \cdot (-1)^l = \operatorname{sgn}(\sigma) \cdot \operatorname{sgn}(\tau).$$

■

Multiplicativity of the Determinant

Theorem 20.b.5 (*Multiplicativity of the Determinant*): For any matrices $A, B \in M_{n\times n}(K)$, we have $\det(A \cdot B) = \det(A) \cdot \det(B)$.

Proof: Fix a matrix $B \in M_{n\times n}(K)$, and consider a function $D : M_{n\times n}(K) \to K$ defined by $D(A) := \det(A \cdot B)$.

Claim: D is n-linear and alternating.

Proof of claim: Let the rows of A be $\alpha_1, \ldots, \alpha_n$. If we view α_i as a $1 \times n$ matrix, then the matrix product $\alpha_i \cdot B$ is also $1 \times n$. In fact, the rows of the product matrix $A \cdot B$ are exactly $\alpha_1 \cdot B, \ldots, \alpha_n \cdot B$. So, $D(A) = \det(\alpha_1 \cdot B, \ldots, \alpha_n \cdot B)$.

Clearly, for two row vectors α_i and α_i', we have $(\alpha_i + \alpha_i') \cdot B = \alpha_i \cdot B + \alpha_i' \cdot B$. Also, for any scalar $c \in K$, $(c\alpha_i) \cdot B = c(\alpha_i \cdot B)$. Since the determinant function det is n-linear with respect to its rows, it heavily follows that D is also n-linear.

Furthermore, if two rows of A are equal (i.e., $\alpha_i = \alpha_j$), then the corresponding rows of $A \cdot B$ are also equal ($\alpha_i \cdot B = \alpha_j \cdot B$). Because det is alternating, $D(A) = \det(\alpha_1 \cdot B, \ldots, \alpha_n \cdot B) = 0$. This rigorously proves the claim that D is alternating.

We continue with the proof of Theorem 20.b.5. By Corollary 20.b.3, any n-linear alternating function satisfies $D(A) = \det(A) \cdot D(I)$. We compute $D(I)$ using our definition: $D(I) = \det(I \cdot B) = \det(B)$. Substituting this back yields $D(A) = \det(A) \cdot \det(B)$, completing the proof. ■

Corollary 20.b.6 (*Determinant of an Inverse*)**:** If A is invertible, then $\det(A) \neq 0$. Moreover, $\det(A^{-1}) = \frac{1}{\det(A)}$.

Proof: Assume A is invertible, and denote its inverse by A^{-1}. Since $A \cdot A^{-1} = I$, taking the determinant of both sides and applying Theorem 20.b.5 yields $\det(A) \cdot \det(A^{-1}) = \det(I) = 1$. This automatically implies that $\det(A)$ cannot be zero, and dividing by $\det(A)$ yields the expected formula. ■

More on Determinants: Properties of Determinants and the Transpose (Section 20.c)

Recall that for any matrix $M \in M_{m\times n}(K)$, we have the transposed matrix $M^T \in M_{n\times m}(K)$, where the entries are given by $M^T(i,j) = M(j,i)$.

Theorem 20.c.1 (*Determinant of the Transpose*)**:** For all $A \in M_{n\times n}(K)$, we have $\det(A^T) = \det(A)$.

Proof: If σ is a permutation of degree n, then by definition of the transpose, $A^T(i,\sigma(i)) = A(\sigma(i),i)$. Therefore, expanding the determinant definition gives:

$$\det(A^T) = \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma)\cdot A(\sigma(1),1)\cdots A(\sigma(n),n).$$

If we let $j = \sigma(i)$, then $i = \sigma^{-1}(j)$, and consequently $A(\sigma(i),i) = A(j,\sigma^{-1}(j))$. Hence, we can rearrange the product completely:

$$A(\sigma(1),1)\cdots A(\sigma(n),n) = A(1,\sigma^{-1}(1))\cdots A(n,\sigma^{-1}(n)) \tag{20.3}$$

This holds because every factor on the left-hand side of (20.3) will appear exactly once on the right-hand side, and vice-versa.

Now, since $\sigma\circ\sigma^{-1} = \mathrm{id}$, we have $\operatorname{sgn}(\sigma)\cdot\operatorname{sgn}(\sigma^{-1}) = \operatorname{sgn}(\mathrm{id}) = 1$, which implies $\operatorname{sgn}(\sigma) = \operatorname{sgn}(\sigma^{-1})$. When σ runs over S_n (all possible permutations of degree n), its inverse σ^{-1} also naturally runs over the entirety of S_n.

This implies:

$$\begin{aligned}\det(A^T) &= \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma)A(\sigma(1),1)\cdots A(\sigma(n),n)\\ &= \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma^{-1})A(1,\sigma^{-1}(1))\cdots A(n,\sigma^{-1}(n))\\ &= \sum_{\tau\in S_n} \operatorname{sgn}(\tau)A(1,\tau(1))\cdots A(n,\tau(n))\\ &= \det(A).\end{aligned}$$

■

Example: Let $A = \left(\begin{smallmatrix}1&2\\3&4\end{smallmatrix}\right)$. Its transpose is $A^T = \left(\begin{smallmatrix}1&3\\2&4\end{smallmatrix}\right)$. We have $\det(A) = (1)(4) - (2)(3) = -2$, and $\det(A^T) = (1)(4) - (3)(2) = -2$. Thus, the theorem holds.

Corollary 20.c.2 (*Column Alternating Property*)**:** If $D : M_{n\times n}(K) \to K$ is an alternating n-linear function of the rows of the matrix, then D is also an alternating n-linear function of the columns of the matrix. In fact, $D(A^T) = D(A)$ for all $A \in M_{n\times n}(K)$. In particular, this holds for $D = \det$.

Proof: For any $A \in M_{n\times n}(K)$, we have previously established $D(A) = \det(A)\cdot D(I)$. Applying this to A^T gives $D(A^T) = \det(A^T)\cdot D(I)$. Since $\det(A^T) = \det(A)$, this evaluates exactly to $\det(A)\cdot D(I) = D(A)$. ■

Because the determinant acts symmetrically on rows and columns, we can perform column operations just as we do row operations.

Theorem 20.c.3 (*Invariance Under Row Operations*): Let $A \in M_{n\times n}(K)$.

(a) If B is obtained from A by adding a multiple of one row of A to another row (i.e., $R_i + cR_j \to R_i$ for $i \neq j$), or by doing the analogous operation on columns, then $\det(B) = \det(A)$.

(b) If B is obtained from A by interchanging two rows (i.e., $R_i \leftrightarrow R_j$ for $i \neq j$), or columns, then $\det(B) = -\det(A)$.

(c) If B is obtained from A by multiplying a row by a scalar (i.e., $cR_i \to R_i$), then $\det(B) = c\det(A)$.

Proof: Statement (b) follows directly from the fact that the determinant is an alternating function. Statement (c) follows from the n-linearity of the determinant.

To prove (a), let A have rows $\alpha_1, \ldots, \alpha_n$. Assume $j < i$. Expanding by linearity on the i-th row gives:

$$\begin{aligned}\det(B) &= \det(\alpha_1, \ldots, \alpha_i + c\alpha_j, \ldots, \alpha_j, \ldots, \alpha_n)\\ &= \det(\alpha_1, \ldots, \alpha_i, \ldots, \alpha_n) + c\det(\alpha_1, \ldots, \alpha_j, \ldots, \alpha_j, \ldots, \alpha_n)\\ &= \det(A) + 0 = \det(A).\end{aligned}$$

The second term identically vanishes because two rows are equal (α_j appears twice). If $j > i$, the proof is similar. ■

Determinants of Block Matrices

Calculating the determinant of a large matrix by expanding all permutations is computationally expensive. Block matrices provide a powerful shortcut. Consider a block matrix M of the form $M = \left(\begin{smallmatrix} A & B \\ 0 & C \end{smallmatrix}\right)$, where $A \in M_{r\times r}(K)$, $C \in M_{s\times s}(K)$, and $B \in M_{r\times s}(K)$.

Proposition 20.c.4 (*Determinant of Block Matrices*): For a block matrix M defined as above, $\det(M) = \det(A)\cdot\det(C)$.

Proof: Define a helper function $D(A, B, C) := \det\left(\begin{smallmatrix} A & B \\ 0 & C \end{smallmatrix}\right)$ as a function $D : M_{r\times r}(K) \times M_{r\times s}(K) \times M_{s\times s}(K) \to K$.

If we fix the first two matrices A and B, then the function mapping $C \mapsto D(A, B, C)$ is s-linear and alternating. By Corollary 20.b.3, any such function satisfies:

$$D(A, B, C) = \det(C)\cdot D(A, B, I_s). \tag{20.4}$$

Let us calculate $D(A, B, I_s) = \det\left(\begin{smallmatrix} A & B \\ 0 & I_s \end{smallmatrix}\right)$. By doing row operations of the type $R_i + cR_j \to R_i$ (where $j \neq i$), we can use the last s rows of the identity matrix I_s to eliminate the entries in B entirely. This transforms the matrix into $\left(\begin{smallmatrix} A & 0 \\ 0 & I_s \end{smallmatrix}\right)$.

Since these operations do not change the determinant, we have:

$$\det\begin{pmatrix} A & B \\ 0 & I_s \end{pmatrix} = \det\begin{pmatrix} A & 0 \\ 0 & I_s \end{pmatrix}. \tag{20.5}$$

The function $A \mapsto \det\left(\begin{smallmatrix} A & 0 \\ 0 & I_s \end{smallmatrix}\right)$ is r-linear and alternating. By Corollary 20.b.3 again, it evaluates to $\det(A)\cdot\det\left(\begin{smallmatrix} I_r & 0 \\ 0 & I_s \end{smallmatrix}\right) = \det(A)\cdot\det(I_{r+s}) = \det(A)\cdot 1 = \det(A)$.

Going back through (20.5) and then to (20.4), we finally get that $D(A, B, C) = \det(A)\cdot\det(C)$. ■

Remark (*Exercise*): Show that $\det\left(\begin{smallmatrix} A & 0 \\ B & C \end{smallmatrix}\right) = \det(A)\cdot\det(C)$, where $A \in M_{r\times r}(K)$, $C \in M_{s\times s}(K)$, and $B \in M_{s\times r}(K)$.

Example (*Calculation of a 4×4 Determinant*): Let $A = \left(\begin{smallmatrix} 1 & -1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 4 & 1 & -1 & -1 \\ 1 & 2 & 3 & 0 \end{smallmatrix}\right) \in M_{4\times 4}(\mathbb{R})$. We begin by performing row operations $R_2 - 2R_1 \to R_2$, $R_3 - 4R_1 \to R_3$, and $R_4 - R_1 \to R_4$. This transforms A into:

$$\begin{pmatrix} 1 & -1 & 2 & 3 \\ 0 & 4 & -4 & -4 \\ 0 & 5 & -9 & -13 \\ 0 & 3 & 1 & -3 \end{pmatrix}$$

Now, we clean the second column below the diagonal using $R_3 - \frac{5}{4}R_2 \to R_3$ and $R_4 - \frac{3}{4}R_2 \to R_4$. This reveals a beautiful block structure:

$$\left(\begin{array}{cc|cc} 1 & -1 & 2 & 3 \\ 0 & 4 & -4 & -4 \\ \hline 0 & 0 & -4 & -8 \\ 0 & 0 & 4 & 0 \end{array}\right)$$

By applying Proposition 20.c.4 to this partitioned form, we can see that the determinant of the 4×4 matrix is simply the product of the determinants of the 2×2 diagonal blocks:

$$\det(A) = \det \begin{pmatrix} 1 & -1 \\ 0 & 4 \end{pmatrix} \cdot \det \begin{pmatrix} -4 & -8 \\ 4 & 0 \end{pmatrix}$$

Consequently, we evaluate the individual pieces:

$$\det(A) = (4) \cdot (32) = 128.$$

Cofactor Expansion and the Classical Adjoint

Recall that if D is a determinant function on $M_{(n-1)\times(n-1)}(K)$, then for any fixed $j \in \{1, \ldots, n\}$, the function $E_j(A) := \sum_{i=1}^{n} (-1)^{i+j} A_{ij} D(A(i|j))$ is a determinant function as well. But, we have seen that on $M_{n\times n}(K)$ there is a strictly unique determinant function, det.

Corollary 20.c.5 (*Laplace Expansion*): For all $A \in M_{n\times n}(K)$, we have the following n different formulas for $\det(A)$ (expanding along column j):

$$\det(A) = \sum_{i=1}^{n} (-1)^{i+j} A_{ij} \det A(i|j), \quad \text{for any } j = 1, \ldots, n.$$

The scalars $C_{ij} := (-1)^{i+j} \det A(i|j)$ are called the i, j *cofactor of* A. We can succinctly write the expansion of $\det(A)$ by the cofactors of the j-th column as $\det(A) = \sum_{i=1}^{n} A_{ij} \cdot C_{ij}$.

Lemma (*Orthogonality of Cofactors*): If we replace A_{ij} in the formula by A_{ik} (where k is fixed), we always get 0. In other words, for all $A \in M_{n\times n}(K)$ and for $j \neq k$, we have $\sum_{i=1}^{n} A_{ik} \cdot C_{ij} = 0$.

Proof: Let $B \in M_{n\times n}(K)$ be the matrix obtained from A by replacing (not exchanging!) column j of A by column k of A. So, in B, column j equals column k. Clearly, $\det(B) = 0$ since B has two equal columns.

Expanding $\det(B)$ along the j-th column gives:

$$0 = \det(B) = \sum_{i=1}^{n} (-1)^{i+j} B_{ij} \det B(i|j).$$

By construction, $B_{ij} = A_{ik}$ for all i. Furthermore, since we only modified column j, deleting column j from both matrices leaves identical submatrices: $B(i|j) = A(i|j)$ for all i. Thus:

$$0 = \sum_{i=1}^{n} (-1)^{i+j} A_{ik} \det A(i|j) = \sum_{i=1}^{n} A_{ik} \cdot C_{ij}.$$

■

Summary: For $A \in M_{n\times n}(K)$, we have $\sum_{i=1}^{n} A_{ik} \cdot C_{ij} = \delta_{jk} \cdot \det(A)$, where δ_{jk} is the Kronecker delta (which is 1 if $j = k$, and 0 otherwise).

Definition (*Classical Adjoint*): The transpose of the matrix of cofactors, $(C_{ij})^T$, is called the *classical adjoint* (or adjugate) of A, denoted $\operatorname{adj} A \in M_{n\times n}(K)$. Explicitly, $(\operatorname{adj} A)_{ij} = C_{ji} = (-1)^{i+j} \det A(j|i)$.

Our summary formula above immediately implies the matrix equation:

$$(\operatorname{adj} A) \cdot A = (\det A) \cdot I \tag{20.6}$$

Lemma (*Left Adjugate Identity*): It is also true that $A \cdot (\operatorname{adj} A) = (\det A) \cdot I$.

Proof: Since $A^T(i|j) = (A(j|i))^T$, we have

$$(\operatorname{adj} A^T)_{ij} = (-1)^{i+j} \det A^T(j|i) = (-1)^{i+j} \det(A(i|j)^T) = (-1)^{i+j} \det A(i|j).$$

This perfectly matches $(\operatorname{adj} A)_{ji}$, which is exactly $((\operatorname{adj} A)^T)_{ij}$. Thus, $\operatorname{adj} A^T = (\operatorname{adj} A)^T$.

Applying (20.6) to the matrix A^T gives $(\operatorname{adj} A^T) \cdot A^T = (\det A^T) \cdot I$. Using our transpose identities, this becomes $(\operatorname{adj} A)^T \cdot A^T = (\det A) \cdot I$. Taking the transpose of both sides finally yields $A \cdot (\operatorname{adj} A) = (\det A) \cdot I$. ■

Theorem 20.c.6 (*Inverse via Adjugate*): Let $A \in M_{n\times n}(K)$. Then, A is invertible if and only if $\det A \neq 0$. When A is invertible, its inverse is given by $A^{-1} = \frac{1}{\det A} \cdot (\operatorname{adj} A)$.

Cramer's Rule

Let $A \in M_{n\times n}(K)$. We would like an explicit formula to solve the system of linear equations $A \cdot x = b$, where $x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ are the unknowns and $b = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} \in K^n$ is a given vector.

Assume $A \cdot x = b$. Multiplying both sides by the adjoint gives $(\operatorname{adj} A) \cdot A \cdot x = (\operatorname{adj} A) \cdot b$. This directly implies $(\det A) \cdot x = (\operatorname{adj} A) \cdot b$. Looking at the j-th row of this vector equation, we get:

$$\begin{aligned}(\det A) \cdot x_j &= \sum_{i=1}^{n} (\operatorname{adj} A)_{ji} \cdot b_i \\ &= \sum_{i=1}^{n} (-1)^{i+j} b_i \det A(i|j) \\ &= \det(A(j,b))\end{aligned}$$

where $A(j,b)$ is the matrix obtained from A by replacing column j with the vector b.

Theorem 20.c.7 (*Cramer's Rule*): Let $A \in M_{n\times n}(K)$ with $\det A \neq 0$. Let $b \in K^n$. Then, the system $Ax = b$ has a unique solution, and it is given by the formula:

$$x_j = \frac{\det A(j,b)}{\det A} \quad \text{for all } 1 \leq j \leq n,$$

where $A(j,b)$ is the matrix obtained from A by replacing column j with b.

Proof of Theorem 20.c.7: We have seen previously in Linear Algebra I that if A is invertible, then $Ax = b$ has a unique solution. We have just shown above that $(\det A) \cdot x_j = \det A(j,b)$. Since $\det A \neq 0$, we can safely divide to obtain $x_j = \frac{\det A(j,b)}{\det A}$. ■

Example: Consider the system $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 3 \end{smallmatrix}\right)\left(\begin{smallmatrix} x_1 \\ x_2 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 5 \\ 5 \end{smallmatrix}\right)$. We have $\det(A) = 2(3) - 1(1) = 5$. Using Cramer's rule:

$$x_1 = \frac{\det \begin{pmatrix} 5 & 1 \\ 5 & 3 \end{pmatrix}}{5} = \frac{15-5}{5} = 2, \quad x_2 = \frac{\det \begin{pmatrix} 2 & 5 \\ 1 & 5 \end{pmatrix}}{5} = \frac{10-5}{5} = 1.$$

Thus, the solution is $x = \left(\begin{smallmatrix} 2 \\ 1 \end{smallmatrix}\right)$.

Determinants of Endomorphisms

Warm up: Let $A \in M_{n\times n}(K)$, and let $P \in GL(n,K)$ (meaning P is an invertible $n \times n$ matrix). Then, by multiplicativity (Theorem 20.b.5):

$$\det(P^{-1} \cdot A \cdot P) = (\det P^{-1}) \cdot (\det A) \cdot (\det P) = \frac{1}{\det P} \cdot \det A \cdot \det P = \det A.$$

In other words, similar matrices mathematically have the exact same determinant.

Let V be a finite-dimensional vector space over K, and put $n := \dim V$. Let $T : V \to V$ be a linear map (an endomorphism). Pick a basis $\mathcal{B}$ for V. We have a matrix representation $[T]_{\mathcal{B}}^{\mathcal{B}} \in M_{n\times n}(K)$, and we can consider its determinant $\det[T]_{\mathcal{B}}^{\mathcal{B}} \in K$.

Question: Does $\det[T]_{\mathcal{B}}^{\mathcal{B}}$ depend on the specific choice of the basis $\mathcal{B}$?

Answer: Let $\mathcal{B}'$ be another arbitrary basis for V. Denote by $P := [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'}$ the transition matrix between the basis $\mathcal{B}'$ and $\mathcal{B}$. Recall that $[T]_{\mathcal{B}'}^{\mathcal{B}'} = [\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{B}} \cdot [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'}$. We also formally have $[\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}} = ([\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'})^{-1} = P^{-1}$. So, $[T]_{\mathcal{B}'}^{\mathcal{B}'} = P^{-1} \cdot [T]_{\mathcal{B}}^{\mathcal{B}} \cdot P$. This implies $\det[T]_{\mathcal{B}'}^{\mathcal{B}'} = \det[T]_{\mathcal{B}}^{\mathcal{B}}$.

Recall also that if $T, S : V \to V$ are two linear maps, then $[S \circ T]_{\mathcal{B}}^{\mathcal{B}} = [S]_{\mathcal{B}}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{B}}$. If T is an isomorphism, then $[T^{-1}]_{\mathcal{B}}^{\mathcal{B}} = ([T]_{\mathcal{B}}^{\mathcal{B}})^{-1}$. Finally, $[\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}} = I$.

Corollary 20.c.8 (*Determinant of an Endomorphism*): Let V be a finite-dimensional vector space over K, and let $T : V \to V$ be a linear map. Then, $\det([T]_{\mathcal{B}}^{\mathcal{B}})$ does NOT depend on the choice of the basis $\mathcal{B}$ of V. We denote this intrinsic scalar simply by $\det(T) \in K$.

The function $\det : \mathrm{End}(V) \to K, T \mapsto \det(T)$ has the following elegant properties:

(a) $\det(S \circ T) = \det(S) \cdot \det(T)$ for all $S, T \in \mathrm{End}(V)$.

(b) T is an isomorphism if and only if $\det T \neq 0$. If T is an isomorphism, then $\det(T^{-1}) = \frac{1}{\det T}$.

(c) $\det(\mathrm{id}_V) = 1$.

(d) $\det(0) = 0 \in K$ (where 0 is the zero linear map).

Important remark: In contrast to an endomorphism $T : V \to V$, if we take a linear map $T : V \to W$ between two vector spaces of the same dimension and bases $\mathcal{B}, \mathcal{C}$ of V, W, then $\det[T]_{\mathcal{C}}^{\mathcal{B}}$ generally depends on the choice of $\mathcal{B}$ and $\mathcal{C}$.

Eigenvalues & Eigenvectors (Chapter 21)

Motivation (Section 21.a)

Fibonacci revisited

Consider the sequence $f_0, f_1, f_2, f_3, \dots, f_n, \dots$ defined by $f_0 = 0$, $f_1 = 1$, and $f_n := f_{n-1} + f_{n-2}$ for $n \geq 2$. This yields the sequence: $0, 1, 1, 2, 3, 5, 8, 13, 21, \dots$

We found a formula for f_n:

$$f_n = \frac{1}{\sqrt{5}}\left(\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right) \quad \forall n \geq 0.$$

We defined $Fib :=$ the set of all sequences $(a_0, a_1, a_2, \dots, a_n, \dots)$ that satisfy $a_n = a_{n-1} + a_{n-2}$ for $n \geq 2$.

Claim: $Fib \subseteq \mathbb{R}^\infty$ is a 2-dimensional vector subspace of the space of all infinite sequences

$$(x_0, x_1, x_2, \dots, x_n, \dots).$$

In fact, the map $F : Fib \longrightarrow \mathbb{R}^2$ given by $F(a_0, a_1, \dots, a_n, \dots) := (a_0, a_1) \in \mathbb{R}^2$ is an isomorphism.

Claim: Put $\varphi = \frac{1+\sqrt{5}}{2}$ and $\Psi := \frac{1-\sqrt{5}}{2}$. Then the two geometric sequences $(1, \varphi, \varphi^2, \dots, \varphi^n, \dots)$ and $(1, \Psi, \Psi^2, \dots, \Psi^n, \dots)$ belong to Fib and, moreover, they form a basis for Fib.

We also saw that using these two sequences, we could easily find an explicit formula for all Fibonacci sequences. But how did we find these two special sequences?

The shift map $S : Fib \longrightarrow Fib$ defined by $S(a_0, a_1, \dots, a_n, \dots) := (a_1, a_2, \dots, a_{n+1}, \dots)$ is a linear map. Let us try to find a Fibonacci sequence $v \neq 0$ such that $S(v) = \lambda \cdot v$ for some λ.

$$S(v) = \lambda \cdot v \iff \lambda = \frac{1 \pm \sqrt{5}}{2} \text{ and } v = (a, a\lambda, a\lambda^2, \dots, a\lambda^n, \dots)$$

Definition 21.a.1 (*Eigenvalue and Eigenvector*): Let V be a vector space over K and $T : V \to V$ a linear map (an endomorphism of V). We say that $\lambda \in K$ is an eigenvalue (a.k.a. characteristic value) of T if there exists a vector $v \neq 0$ such that $Tv = \lambda v$. Any vector $v \neq 0$ with the property that $Tv = \lambda v$ is called an eigenvector (of T) for the eigenvalue λ.

Definition 21.a.2 (*Diagonalizable Endomorphism*): Let V be a finite-dimensional vector space over K and $T : V \to V$ a linear map. We say that T is diagonalizable if there exists a basis $\mathcal{B}$ of V such that

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} \lambda_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_n \end{pmatrix} \tag{21.1}$$

Lemma 21.a.3 (*Diagonalizable Matrix Criterion*): T is diagonalizable if and only if there exists a basis $\mathcal{B} = (v_1, \dots, v_n)$ consisting of eigenvectors of T (i.e., $\forall i$, $Tv_i = \lambda_i \cdot v_i$ for some $\lambda_i \in K$). Moreover, if for some basis $\mathcal{B} = (v_1, \dots, v_n)$ the matrix $[T]_{\mathcal{B}}^{\mathcal{B}}$ has the shape (21.1), then for all i, λ_i is an eigenvalue of T and v_i is an eigenvector for λ_i.

Proof: Recall that $[T]_{\mathcal{B}}^{\mathcal{B}} = ([Tv_1]_{\mathcal{B}} \dots [Tv_n]_{\mathcal{B}})$ and $[Tv_j]_{\mathcal{B}} = \begin{pmatrix} \alpha_{1j} \\ \vdots \\ \alpha_{nj} \end{pmatrix}$, where $Tv_j = \alpha_{1j}v_1 + \dots + \alpha_{nj}v_n$. The matrix $[T]_{\mathcal{B}}^{\mathcal{B}}$ has the shape (21.1) if and only if $\alpha_{ij} = 0$ for all $i \neq j$, and $\alpha_{kk} = \lambda_k \iff Tv_k = \lambda_k \cdot v_k$. ■

Remark: We saw that for all $A \in M_{n\times n}(K)$, we have a linear map $T_A : K^n \to K^n$, $T_A(v) := A \cdot v$. We say that a matrix $A \in M_{n\times n}(K)$ is diagonalizable if the linear map T_A is diagonalizable.

Example: Let $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ and $K = \mathbb{R}$. $T_A\left(\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 3 \\ 3 \end{smallmatrix}\right) = 3 \cdot \left(\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right)$. Thus, $\lambda = 3$ is an eigenvalue and $v = \left(\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right)$ is a corresponding eigenvector. Furthermore, $T_A\left(\begin{smallmatrix} 1 \\ -1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 1 \\ -1 \end{smallmatrix}\right)$. This implies that $\lambda = 1$ is an eigenvalue and $v = \left(\begin{smallmatrix} 1 \\ -1 \end{smallmatrix}\right)$ is a corresponding eigenvector. Geometrically, this represents a stretching by a factor of 3 in $\mathbb{R}^2$ along one axis and the identity map along the other axis.

Example: Let $A = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$. Let us find all possible eigenvalues and eigenvectors. We are looking for $A \cdot v = \lambda v$ with $\lambda \in \mathbb{R}$ and $0 \neq v \in \mathbb{R}^2$. This gives $(A - \lambda I)v = 0$. This implies that $A - \lambda I$ is not invertible, which in turn means $\det(A - \lambda I) = 0$ (which leads to a quadratic equation).

Proposition 21.a.4 (*Linear Independence of Eigenvectors*): Let V be a vector space over K and $T : V \to V$ a linear map. Suppose that $\lambda_1, \ldots, \lambda_n \in K$ are eigenvalues of T which are pairwise distinct (i.e., $\lambda_i \neq \lambda_j$ for all $i \neq j$). Let $v_1, \ldots, v_n$ be eigenvectors for $\lambda_1, \ldots, \lambda_n$ respectively. Then $v_1, \ldots, v_n$ are linearly independent. (Put shortly: eigenvectors corresponding to a list of distinct eigenvalues are always linearly independent).

Proof: We proceed by induction on n. Base case $n = 1$: v_1 is an eigenvector for λ_1. By Definition 21.a.1, $v_1 \neq 0$, which implies that v_1 is linearly independent.

Inductive step for $n \geq 1$: Suppose the proposition holds for any list of n eigenvalues and eigenvectors. We will prove now that the proposition holds also for a list of $n + 1$ eigenvalues and eigenvectors. Let $\lambda_1, \ldots, \lambda_n, \lambda_{n+1}$ be pairwise distinct eigenvalues of T, and $v_1, \ldots, v_n, v_{n+1}$ a given list of eigenvectors of $\lambda_1, \ldots, \lambda_n, \lambda_{n+1}$ respectively. We need to prove that $v_1, \ldots, v_n, v_{n+1}$ are linearly independent. Assume that

$$a_1 v_1 + \cdots + a_n v_n + a_{n+1} v_{n+1} = 0. \tag{21.2}$$

We need to show that $a_1 = \cdots = a_n = a_{n+1} = 0$. Apply the transformation T to (21.2):

$$a_1 T(v_1) + \cdots + a_n T(v_n) + a_{n+1} T(v_{n+1}) = 0.$$

But since $T(v_i) = \lambda_i v_i$, we have:

$$a_1 \lambda_1 v_1 + \cdots + a_n \lambda_n v_n + a_{n+1} \lambda_{n+1} v_{n+1} = 0. \tag{21.3}$$

Consider now the combination (21.3) $- \lambda_{n+1} \cdot$ (21.2):

$$a_1(\lambda_1 - \lambda_{n+1})v_1 + \cdots + a_n(\lambda_n - \lambda_{n+1})v_n + a_{n+1}(\lambda_{n+1} - \lambda_{n+1})v_{n+1} = 0.$$

By the induction hypothesis, $v_1, \ldots, v_n$ are linearly independent. Therefore, $a_1(\lambda_1 - \lambda_{n+1}) = \cdots = a_n(\lambda_n - \lambda_{n+1}) = 0$. By assumption, the eigenvalues are distinct, so $\lambda_i \neq \lambda_{n+1}$. Consequently, $a_1 = \cdots = a_n = 0$. Going back to (21.2), we get $a_{n+1} v_{n+1} = 0$. But since $v_{n+1} \neq 0$ (because v_{n+1} is an eigenvector), this implies that $a_{n+1} = 0$. ■

The Characteristic Polynomial

Proposition 21.a.5 (*Eigenvalue Injective Criterion*): Let V be a vector space over K and $T : V \to V$ a linear map. Then λ is an eigenvalue of T if and only if the linear map $T - \lambda \cdot \mathrm{id}_V$ is not injective.

Proof: λ is an eigenvalue of $T \iff \exists 0 \neq v \in V$ such that $Tv = \lambda v \iff \exists 0 \neq v \in V$ such that $Tv - \lambda v = 0 \iff \exists 0 \neq v \in V$ such that $(T - \lambda \cdot \mathrm{id}_V)v = 0 \iff \ker(T - \lambda \cdot \mathrm{id}_V) \neq \{0\}$. This implies that $T - \lambda \cdot \mathrm{id}_V$ is not injective. ■

Definition 21.a.6 (*Characteristic Polynomial*): Let V be a finite-dimensional vector space over K and $T : V \to V$ a linear map. We define the characteristic polynomial of T as $P_T(x) := \det(T - x \cdot \mathrm{id}_V)$ (sometimes denoted $\chi_T(x)$), where x is a formal variable ("in K"). We define the characteristic polynomial of a matrix $A \in M_{n \times n}(K)$ in the same way: $P_A(x) := \det(A - x \cdot I_n)$.

Definition 21.a.7 (*Eigenspace*): Let V be a vector space over K and $T \in \mathrm{End}(V)$. Let λ be an eigenvalue of T. The eigenspace corresponding to λ is defined as:

$$\mathrm{Eig}_T(\lambda) := \{v \in V \mid Tv = \lambda v\} = \ker(T - \lambda \cdot \mathrm{id}_V) = \{v \in V \mid v \text{ is an eigenvector of } \lambda\} \cup \{0\}.$$

Clearly (by this definition), $\mathrm{Eig}_T(\lambda) \subseteq V$ is a linear subspace.

Lemma 21.a.8 (*Properties of the Characteristic Polynomial*)**:** Let $A \in M_{n\times n}(K)$. Then:

(a) $P_A(x)$ is a polynomial of degree n with leading coefficient $(-1)^n$ (i.e., $P_A(x) = c_n x^n + c_{n-1}x^{n-1} + \cdots + c_1 x + c_0$, with $c_i \in K$ for all i, and $c_n = (-1)^n$).

(b) $P_{A^T}(x) = P_A(x)$.

(c) If $A = \begin{pmatrix} \lambda_1 & \dots & * \\ 0 & \ddots & \vdots \\ 0 & \dots & \lambda_n \end{pmatrix}$ has upper triangular form, then $P_A(x) = (\lambda_1 - x)\cdot\dots\cdot(\lambda_n - x)$.

(d) If $A = \left(\begin{array}{c|c} B_1 & * \\ \hline 0 & B_2 \end{array}\right)$ is a block matrix, then $P_A(x) = P_{B_1}(x)\cdot P_{B_2}(x)$.

(e) If A and B are similar matrices (i.e., $B = P^{-1}AP$ for some invertible matrix P), then $P_A(x) = P_B(x)$.

Proof: **(a)** Let

$$P_A(x) = \det(A - x\cdot I_n) = \det\begin{pmatrix} A_{11}-x & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22}-x & \dots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{n1} & A_{n2} & \dots & A_{nn}-x \end{pmatrix}.$$

Denote this matrix (inside the determinant) by B. This determinant equals $\sum_{\sigma\in S_n}\operatorname{sgn}(\sigma)\cdot B_{1,\sigma 1}\dots B_{n,\sigma n}$. $= (A_{11}-x)\cdot(A_{22}-x)\dots(A_{nn}-x)+$(products of degree $\le n-1$). If $\sigma \neq \mathrm{id} = \left(\begin{smallmatrix} 1 & 2 & \dots & n \\ 1 & 2 & \dots & n \end{smallmatrix}\right)$, then at least one factor in the product $B_{1,\sigma 1}\dots B_{n,\sigma n}$ does not contain x. $= (-1)^n x^n +$(terms of degree $\le n-1$).

(b) $P_{A^T}(x) = \det(A^T - x\cdot I_n) = \det((A - x\cdot I_n)^T) = \det(A - x\cdot I_n) = P_A(x)$.

(c)+(d) Exercise: Follows from properties of the determinant we have already proved.

(e) $P_B(x) = \det(B - x\cdot I_n) = \det(P^{-1}AP - x\cdot I_n) = \det(P^{-1}(A - x\cdot I_n)P) = \det(A - x\cdot I_n) = P_A(x)$. ■

Remark: If V is an n-dimensional vector space over K and $T \in \operatorname{End}(V)$, then $P_T(x)$ is a polynomial of degree n with leading coefficient $(-1)^n$.

Proof: Let $\mathcal{B}$ be a basis for V. Then $[T - x\cdot \mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}} = [T]_{\mathcal{B}}^{\mathcal{B}} - x\cdot I_n$. This implies that $P_T(x) = \det([T - x\cdot\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}}) = \det([T]_{\mathcal{B}}^{\mathcal{B}} - x\cdot I_n) = P_{[T]_{\mathcal{B}}^{\mathcal{B}}}(x)$. The claim of the remark now follows from **(a)** of Lemma 21.a.8. ■

Definition 21.a.9 (*Trace*)**:** Let $A \in M_{n\times n}(K)$. Define the trace (German: "Spur") of A, denoted $\operatorname{Tr}(A) := \sum_{i=1}^n A_{ii} \in K$ (which is the sum of the entries along the diagonal of A). If V is a finite-dimensional vector space over K and $T \in \operatorname{End}(V)$, define $\operatorname{Tr}(T) := \operatorname{Tr}([T]_{\mathcal{B}}^{\mathcal{B}})$, where $\mathcal{B}$ is some basis for V. We will see soon that $\operatorname{Tr}(T)$ is well defined (i.e., it does not depend on the choice of $\mathcal{B}$).

Lemma 21.a.10 (*Properties of Trace*)**:** For all $A, B \in M_{n\times n}(K)$, we have:

(a) $\operatorname{Tr}(A\cdot B) = \operatorname{Tr}(B\cdot A)$

(b) If A and B are similar matrices, then $\operatorname{Tr}(A) = \operatorname{Tr}(B)$.

In particular, if $\mathcal{B}$ and $\mathcal{B}'$ are two bases for V, then $\operatorname{Tr}([T]_{\mathcal{B}}^{\mathcal{B}}) = \operatorname{Tr}([T]_{\mathcal{B}'}^{\mathcal{B}'})$. Thus, $\operatorname{Tr}(T)$ is well defined.

Proof: **(a)** $\operatorname{Tr}(A\cdot B) = \sum_{i=1}^n (A\cdot B)_{ii} = \sum_{i=1}^n\sum_{j=1}^n A_{ij}\cdot B_{ji} = \sum_{i,j} A_{ij}B_{ji} = \sum_{i,j} A_{ji}B_{ij} = \sum_{i=1}^n\sum_{j=1}^n B_{ij}A_{ji} = \sum_{i=1}^n (B\cdot A)_{ii} = \operatorname{Tr}(B\cdot A)$.

(b) If $B = P^{-1}AP$, then $\operatorname{Tr}(B) = \operatorname{Tr}(P^{-1}AP) = \operatorname{Tr}(P^{-1}\cdot(AP)) = \operatorname{Tr}(AP\cdot P^{-1}) = \operatorname{Tr}(A)$.

Regarding the claim about $\operatorname{Tr}(T)$, recall that $[T]_{\mathcal{B}'}^{\mathcal{B}'} = [\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}}\cdot[T]_{\mathcal{B}}^{\mathcal{B}}\cdot[\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'}$, and $[\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}} = ([\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'})^{-1}$. Use **(b)** to conclude the proof. ■

Why is the trace interesting? Let $A \in M_{n\times n}(K)$ and $P_A(x) = c_n x^n + c_{n-1}x^{n-1} + \cdots + c_1 x + c_0$ its characteristic polynomial. We have seen that $c_n = (-1)^n$.

Lemma 21.a.11 (*Coefficients of the Characteristic Polynomial*)**:** $c_{n-1} = (-1)^{n-1} \cdot \mathrm{Tr}(A)$ and $c_0 = \det(A)$. In other words, $P_A(x) = (-1)^n x^n + (-1)^{n-1}\,\mathrm{Tr}(A) \cdot x^{n-1} + c_{n-2}x^{n-2} + \cdots + c_1 x + \det(A)$.

Proof: $P_A(x) = \det \begin{pmatrix} A_{11} - x & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22} - x & \dots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{n1} & A_{n2} & \dots & A_{nn} - x \end{pmatrix}$. Denote this matrix by B.

$$P_A(x) = (A_{11} - x) \cdot (A_{22} - x) \dots (A_{nn} - x) + \sum_{\sigma \in S_n, \sigma \neq \mathrm{id}} \mathrm{sgn}(\sigma) B_{1,\sigma 1} \dots B_{n,\sigma n}. \tag{21.4}$$

If $\sigma \neq \mathrm{id}$, then there exist at least two indices $i \neq j$ such that $i \neq \sigma i$ and $j \neq \sigma j$. Thus, the product contains at least two factors ($B_{i,\sigma i}$ and $B_{j,\sigma j}$) that are not from the diagonal of B, and so do not contain x. Therefore, the degree of the sum in (21.4) is $\leq n-2$. $P_A(x) = (A_{11} - x)(A_{22} - x) \dots (A_{nn} - x) + (\text{another polynomial of degree} \leq n-2)$. $= (-1)^n x^n + \left[A_{11}(-x)^{n-1} + \cdots + A_{nn}(-x)^{n-1}\right] + \dots$ $= (-1)^n x^n + (-1)^{n-1} \cdot \mathrm{Tr}(A) x^{n-1} + \cdots + c_1 x + c_0$.

Finally, $c_0 = P_A(0) = \det(A - 0 \cdot I_n) = \det(A)$ (obtained by substituting $x = 0$). ■

Definition 21.a.12 (*Direct Sum*)**:** Let V be a vector space over K and $U_1, \dots, U_r \subseteq V$ subspaces. Denote $W := U_1 + \cdots + U_r \subseteq V$. We say that W is a direct sum of $U_1, \dots, U_r$ (or that $U_1 + \cdots + U_r$ is a direct sum) if every $w \in W$ has a unique representation as $w = u_1 + \cdots + u_r$, with $u_i \in U_i$ for all i.

Exercise: $U_1 + \cdots + U_r$ is a direct sum if and only if any of the following conditions hold:

(a) The only way to write $0 = u_1 + \cdots + u_r$, with $u_i \in U_i$ for all i, is with $u_1 = \cdots = u_r = 0$.

(b) For all $2 \leq j \leq r$, we have $U_j \cap (U_1 + \cdots + U_{j-1}) = \{0\}$.

Under the additional assumption that $U_1, \dots, U_r$ are all finite-dimensional, $U_1 + \cdots + U_r$ is a direct sum if and only if whenever $\mathcal{B}_1, \dots, \mathcal{B}_r$ are bases for $U_1, \dots, U_r$ respectively, then $(\mathcal{B}_1, \dots, \mathcal{B}_r)$ forms a basis for $U_1 + \cdots + U_r$.

Exercise: If $U_1 + \cdots + U_r$ is a direct sum, then there exists a canonical isomorphism $U_1 \oplus \cdots \oplus U_r \xrightarrow{\sim} U_1 + \cdots + U_r$, given by $(u_1, \dots, u_r) \mapsto u_1 + \cdots + u_r$. So when $U_1 + \cdots + U_r$ is a direct sum, we can identify $U_1 + \cdots + U_r$ with $U_1 \oplus \cdots \oplus U_r$.

Proposition 21.a.13 (*Direct Sum of Eigenspaces*)**:** Let V be a vector space over K and $T \in \mathrm{End}(V)$. Let $\lambda_1, \dots, \lambda_r \in K$ be pairwise distinct eigenvalues of T. Then $\mathrm{Eig}_T(\lambda_1) + \cdots + \mathrm{Eig}_T(\lambda_r)$ is a direct sum.

Proof: Assume $u_1 + \cdots + u_r = u_1' + \cdots + u_r'$, where $u_i, u_i' \in \mathrm{Eig}_T(\lambda_i)$ for all $1 \leq i \leq r$. We need to show that $u_i = u_i'$ for all $i \iff (u_i - u_i' = 0$ for all $i)$. We have:

$$(u_1 - u_1') + \cdots + (u_r - u_r') = 0. \tag{21.5}$$

Assume by contradiction that there exists an i such that $u_i - u_i' \neq 0$. Remove from the list $u_1 - u_1', \dots, u_r - u_r'$ all those vectors that are 0. We get a non-empty list $u_{i_1} - u_{i_1}', \dots, u_{i_\ell} - u_{i_\ell}'$ of non-zero vectors, and by (21.5), we still have

$$(u_{i_1} - u_{i_1}') + \cdots + (u_{i_\ell} - u_{i_\ell}') = 0. \tag{21.6}$$

Now for all k, $u_{i_k} - u_{i_k}'$ is either an eigenvector of λ_{i_k} or 0, but by our assumptions, $u_{i_k} - u_{i_k}' \neq 0$. Recall from Proposition 21.a.4 that eigenvectors corresponding to pairwise distinct eigenvalues are linearly independent. $u_{i_1} - u_{i_1}', \dots, u_{i_\ell} - u_{i_\ell}'$ are linearly independent vectors, and by (21.6), their sum is 0. This is a contradiction. ■

Corollary 21.a.14 (*Diagonalizability via Eigenspaces*)**:** Assume V is finite-dimensional, $T \in \mathrm{End}(V)$, and let $\lambda_1, \dots, \lambda_r$ be all the eigenvalues of T. Then T is diagonalizable if and only if $\sum_{i=1}^r \dim \mathrm{Eig}_T(\lambda_i) = \dim V$.

Exercise: If V is finite-dimensional, then T can have only finitely many eigenvalues. (Of course, it is implicitly assumed $\lambda_j \neq \lambda_k$ for all $j \neq k$).

Proof: Choose a basis $\mathcal{B}_i$ for $\mathrm{Eig}_T(\lambda_i)$. Recall that $W := \mathrm{Eig}_T(\lambda_1) + \cdots + \mathrm{Eig}_T(\lambda_r)$ is a direct sum. This implies that $(\mathcal{B}_1, \ldots, \mathcal{B}_r)$ is a basis for W. Now $W \subseteq V$ is a linear subspace, and $\dim W = \sum_{i=1}^r |\mathcal{B}_i| = \sum_{i=1}^r \dim \mathrm{Eig}_T(\lambda_i) \leq \dim V$. This implies $W = V$ if and only if $\sum_{i=1}^r \dim \mathrm{Eig}_T(\lambda_i) = \dim V$. Consequently, V has a basis (namely $(\mathcal{B}_1, \ldots, \mathcal{B}_r)$) whose elements are eigenvectors of T if and only if this dimension equality holds. This implies that T is diagonalizable. ■

Multiplicities of Eigenvalues (Section 21.b)

Example: Consider the matrices $A = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$. We have that their characteristic polynomials are identical: $P_A(x) = P_B(x) = (1-x)^2 = (x-1)^2$.

Example (*Jordan Block*): Consider the matrix $J_{\lambda,n} \in M_{n\times n}(K)$, for $\lambda \in K$.

$$J := J_{\lambda,n} := \begin{pmatrix} \lambda & 1 & 0 & \ldots & 0 \\ 0 & \lambda & 1 & \ldots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \lambda & 1 \\ 0 & 0 & 0 & 0 & \lambda \end{pmatrix} \tag{21.7}$$

This is called a Jordan block. It features λ's along the main diagonal, 1's along the "diagonal" just above the main diagonal, and 0's everywhere else. The characteristic polynomial is $P_{J_{\lambda,n}}(x) = (\lambda - x)^n$. This implies that J has only one eigenvalue, namely λ.

Let us find the eigenvectors for the eigenvalue λ:

$$J - \lambda \cdot I_n = \begin{pmatrix} 0 & 1 & 0 & \ldots & 0 \\ 0 & 0 & 1 & \ldots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

This matrix clearly has $n-1$ linearly independent columns. Therefore, the rank is $\mathrm{rank}(J - \lambda \cdot I_n) = n-1$. By the rank-nullity theorem, $\dim \ker(J - \lambda \cdot I_n) = 1$. Consequently, the eigenspace $\mathrm{Eig}_J(\lambda)$ is 1-dimensional. Clearly, $e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$ is an eigenvector for λ. So $\mathrm{Eig}_J(\lambda) = \mathrm{Sp}(e_1)$.

Important remark: There is an important difference between polynomials $p(x) \in K[x]$ and the functions they define $K \to K$. For example, if $K = \mathbb{Z}_2$, then $p(x) = x$ and $q(x) = x^2$ are different polynomials, but they define the same function $\mathbb{Z}_2 \to \mathbb{Z}_2$. (What happens for $K = \mathbb{Q}, \mathbb{R}, \mathbb{C}, \ldots$?)

Lemma 21.b.1 (*Decomposition of Diagonalizable Matrices*): Let $A \in M_{n\times n}(K)$ (or $T \in \mathrm{End}(V)$) be diagonalizable. Then $P_A(x)$ decomposes as a product of linear (i.e., degree 1) factors:

$$P_A(x) = (\lambda_1 - x) \cdot \cdots \cdot (\lambda_n - x),$$

where $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of A (but there might be repetitions among the λ_i's).

Proof: By assumption, there exists an invertible matrix P such that:

$$P^{-1}AP = \begin{pmatrix} \lambda_1 & \ldots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \ldots & \lambda_n \end{pmatrix} =: D. \tag{21.8}$$

(i.e., A is similar to D). Since similar matrices have the same characteristic polynomial, we have:

$$P_A(x) = P_D(x) = \det \begin{pmatrix} \lambda_1 - x & \ldots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \ldots & \lambda_n - x \end{pmatrix} = (\lambda_1 - x) \ldots (\lambda_n - x).$$

■

Example: Let $A = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ over $K = \mathbb{R}$. The associated linear map $T_A : \mathbb{R}^2 \to \mathbb{R}^2$ represents a counter-clockwise rotation. The characteristic polynomial is $P_A(x) = \det\left(\begin{smallmatrix} -x & -1 \\ 1 & -x \end{smallmatrix}\right) = x^2 + 1$. This polynomial has no roots in $\mathbb{R}$. And, of course, A has no real eigenvalues.

A quick digression about polynomials...

Let $0 \neq p(x) \in K[x]$ and $\lambda \in K$. By dividing $p(x)$ by $(x - \lambda)$, we can write:

$$p(x) = (x - \lambda) \cdot q(x) + R, \tag{21.9}$$

where $\deg q(x) = \deg p(x) - 1$, and $R \in K$ is the remainder. If we substitute $x = \lambda$ into (21.9), we obtain $p(\lambda) = 0 \cdot q(\lambda) + R = R$. Thus, $p(x) = (x - \lambda) \cdot q(x) + p(\lambda)$. Conclusion: $p(\lambda) = 0 \iff p(x) = (x - \lambda) \cdot q(x)$ for some $q(x) \in K[x]$. We say that λ is a **zero** of $p(x)$ (De: Nullstelle), or a **root** of $p(x)$. Sometimes, λ is also a zero of $q(x)$, so $q(x) = (x - \lambda) \cdot h(x)$, which implies $p(x) = (x - \lambda)^2 \cdot h(x)$, etc.

Definition 21.b.2 (*Multiplicity of a Zero*): We say that λ is a zero of **multiplicity** $m \in \mathbb{N}_{>0}$ if $p(x) = (x - \lambda)^m \cdot g(x)$, where $g(x) \in K[x]$ and $g(\lambda) \neq 0$ (i.e., λ is not a zero of $g(x)$). If $m = 1$, we say λ is a **simple zero** of $p(x)$. If $m \geq 2$, we say λ is a **multiple zero** of $p(x)$.

Geometric & Algebraic Multiplicity of Eigenvalues

Definition 21.b.3 (*Geometric and Algebraic Multiplicity*): Let V be a vector space over K and $T \in \mathrm{End}(V)$. Let λ be an eigenvalue of T. We define the **geometric multiplicity** of λ to be:

$$m_g(T; \lambda) := \dim \mathrm{Eig}_T(\lambda).$$

Assume now, in addition, that V is finite-dimensional. We define the **algebraic multiplicity** of λ to be the multiplicity of the zero λ in the characteristic polynomial $P_T(x)$ of T. We denote it by $m_a(T; \lambda)$.

Proposition 21.b.4 (*Multiplicity Inequality*): For any eigenvalue λ, the geometric multiplicity is less than or equal to the algebraic multiplicity:

$$m_g(T; \lambda) \leq m_a(T; \lambda).$$

Proof: Let $v_1, \ldots, v_k$ be a basis for the eigenspace $\mathrm{Eig}_T(\lambda)$. We extend this basis to a basis $\mathcal{B} = (v_1, \ldots, v_k, w_{k+1}, \ldots, w_n)$ of V. Then the matrix of T with respect to the basis $\mathcal{B}$ has the block form:

$$[T]_{\mathcal{B}}^{\mathcal{B}} = ([Tv_1]_{\mathcal{B}} \ldots [Tv_k]_{\mathcal{B}} \mid [Tw_{k+1}]_{\mathcal{B}} \ldots [Tw_n]_{\mathcal{B}})$$

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \left(\begin{array}{c|c} \lambda I_k & C_1 \\ \hline 0 & C \end{array}\right) \tag{21.10}$$

where $C \in M_{(n-k)\times(n-k)}(K)$. We compute the characteristic polynomial by utilizing the determinant property for block matrices (as established in previous sections):

$$P_T(x) = \det([T]_{\mathcal{B}}^{\mathcal{B}} - x \cdot I_n) = (\lambda - x)^k \cdot \det(C - xI_{n-k})$$

$$P_T(x) = (\lambda - x)^k \cdot P_C(x) = (-1)^k (x - \lambda)^k \cdot P_C(x).$$

This factorization directly shows that the multiplicity of λ as a zero of $P_T(x)$ is at least k. Therefore, we conclude that $m_a(T; \lambda) \geq k = m_g(T; \lambda)$. ■

Example: Revisiting the Jordan block $J := J_{\lambda,n}$ defined in (21.7): We previously saw that $P_J(x) = (\lambda - x)^n$, which directly implies that $m_a(J; \lambda) = n$. We also found that $\mathrm{Eig}_J(\lambda)$ is a 1-dimensional space, which means $m_g(J; \lambda) = 1$.

Diagonalizability (Section 21.c)

Theorem 21.c.1 (*Equivalent Conditions for Diagonalizability*): Let V be an n-dimensional vector space over K and $T \in \mathrm{End}(V)$. The following statements are equivalent:

(a) T is diagonalizable.

(b) There exists a basis for V whose elements are eigenvectors of T.

(c) The characteristic polynomial $P_T(x)$ splits into a product of linear factors and, for every eigenvalue of T, the geometric and algebraic multiplicities are equal.

(d) $\sum_{i=1}^{k} \dim \mathrm{Eig}_T(\lambda_i) = \dim V$, where $\lambda_1, \ldots, \lambda_k \in K$ are all the distinct eigenvalues of T.

(e) $V \cong \bigoplus_{i=1}^{k} \mathrm{Eig}_T(\lambda_i)$.

Proof: **(a)** $\iff$ **(b)** has already been proved.

(a) $\implies$ **(c)**: Let $\mathcal{B}$ be a basis in which $[T]_{\mathcal{B}}^{\mathcal{B}}$ has diagonal form. By changing the order of the elements in $\mathcal{B}$, we can arrange that $[T]_{\mathcal{B}}^{\mathcal{B}}$ has the following block-diagonal shape:

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \left(\begin{array}{ccc|c|ccc} \lambda_1 & \ldots & 0 & & & & \\ \vdots & \ddots & \vdots & 0 & & 0 & \\ 0 & \ldots & \lambda_1 & & & & \\ \hline & 0 & & \ddots & & 0 & \\ \hline & & & & \lambda_k & \ldots & 0 \\ & 0 & & 0 & \vdots & \ddots & \vdots \\ & & & & 0 & \ldots & \lambda_k \end{array}\right) \tag{21.11}$$

where $\mathcal{B} = (v_1^{(1)}, \ldots, v_{\ell_1}^{(1)}, v_1^{(2)}, \ldots, v_{\ell_2}^{(2)}, \ldots, v_1^{(k)}, \ldots, v_{\ell_k}^{(k)})$. Here $\lambda_1, \ldots, \lambda_k$ are the eigenvalues of T (with $\lambda_i \neq \lambda_j$ for all $i \neq j$) and $v_1^{(i)}, \ldots, v_{\ell_i}^{(i)}$ are eigenvectors of λ_i for all i. The characteristic polynomial is:

$$P_T(x) = \det([T]_{\mathcal{B}}^{\mathcal{B}} - x \cdot I_n) = (\lambda_1 - x)^{\ell_1} \cdot \cdots \cdot (\lambda_k - x)^{\ell_k}. \tag{21.12}$$

This shows that $P_T(x)$ splits as a product of linear factors. It also shows that $\lambda_1, \ldots, \lambda_k$ are all the eigenvalues of T. It remains to show that $m_g(T; \lambda_i) = m_a(T; \lambda_i)$ for all i. Clearly, $v_1^{(i)}, \ldots, v_{\ell_i}^{(i)} \in \mathrm{Eig}_T(\lambda_i)$ are linearly independent; hence $\ell_i \leq \dim \mathrm{Eig}_T(\lambda_i) = m_g(T; \lambda_i)$ by definition. But from (21.12), we also have $\ell_i = m_a(T; \lambda_i)$. So $m_a(T; \lambda_i) \leq m_g(T; \lambda_i)$ for all i. At the same time, we always have the inequality $m_g(T; \lambda_i) \leq m_a(T; \lambda_i)$. Therefore, $m_g(T; \lambda_i) = m_a(T; \lambda_i)$ for all i. (The proof also shows that $\ell_i = m_g(T; \lambda_i) = m_a(T; \lambda_i)$).

(c) $\implies$ **(d)**: We assume that $P_T(x) = (\lambda_1 - x)^{\ell_1} \ldots (\lambda_k - x)^{\ell_k}$ and $m_g(T; \lambda_i) = m_a(T; \lambda_i)$. Then $\lambda_1, \ldots, \lambda_k$ are exactly the eigenvalues of T. By definition, $\ell_i = m_a(T; \lambda_i)$. We have $n = \dim V = \sum_{i=1}^{k} m_a(T; \lambda_i)$ (because $\deg P_T(x) = n$). By assumption, this equals $\sum_{i=1}^{k} m_g(T; \lambda_i) = \sum_{i=1}^{k} \dim \mathrm{Eig}_T(\lambda_i)$.

(d) $\implies$ **(e)**: Recall that $\mathrm{Eig}_T(\lambda_1) + \cdots + \mathrm{Eig}_T(\lambda_k)$ is a direct sum. Consequently, $\dim(\mathrm{Eig}_T(\lambda_1) + \cdots + \mathrm{Eig}_T(\lambda_k)) = \dim \mathrm{Eig}_T(\lambda_1) + \cdots + \dim \mathrm{Eig}_T(\lambda_k) = \dim V$ by assumption. Since $\mathrm{Eig}_T(\lambda_1) + \cdots + \mathrm{Eig}_T(\lambda_k) \subseteq V$ is a linear subspace, we have $\mathrm{Eig}_T(\lambda_1) + \cdots + \mathrm{Eig}_T(\lambda_k) = V$. This implies that $\mathrm{Eig}_T(\lambda_1) + \cdots + \mathrm{Eig}_T(\lambda_k) \cong \mathrm{Eig}_T(\lambda_1) \oplus \cdots \oplus \mathrm{Eig}_T(\lambda_k)$.

(e) $\implies$ **(a)**: If $V \cong \mathrm{Eig}_T(\lambda_1) \oplus \cdots \oplus \mathrm{Eig}_T(\lambda_k)$, then since $\mathrm{Eig}_T(\lambda_1) + \cdots + \mathrm{Eig}_T(\lambda_k)$ is a direct sum and also a linear subspace of V, we must have $\mathrm{Eig}_T(\lambda_1) + \cdots + \mathrm{Eig}_T(\lambda_k) = V$. If we choose a basis $v_1^{(i)}, \ldots, v_{\ell_i}^{(i)}$ for $\mathrm{Eig}_T(\lambda_i)$ for all i (where $\ell_i = m_g(T; \lambda_i)$), then the union $(v_1^{(1)}, \ldots, v_{\ell_1}^{(1)}, \ldots, v_1^{(k)}, \ldots, v_{\ell_k}^{(k)})$ is a basis of V and all the vectors in it are eigenvectors of T. ■

Corollary 21.c.2 (*Sufficient Condition for Diagonalizability*): If $P_T(x)$ splits as a product of linear factors and each eigenvalue λ of T has algebraic multiplicity 1, then T is diagonalizable.

Proof: If $m_a(T; \lambda) = 1$, this implies that $m_g(T; \lambda) = 1$. Now use point **(c)** of Theorem 21.c.1. ■

Trigonalization

(= bringing a matrix/endomorphism to upper triangular form)

Question: Let $T \in \mathrm{End}(V)$. We want to find a basis $\mathcal{B}$ for T such that $[T]_{\mathcal{B}}^{\mathcal{B}}$ has as simple a form as possible.

- If T is diagonalizable, we are okay. But we have seen that this is not always the case. For example, $A = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ is not diagonalizable (we saw: $m_a(A;1) = 2, m_g(A;1) = 1$).
- A weaker (hence more general) condition is trigonalizability.

Definition 21.c.3 (*Trigonalizable Endomorphism*): We say that $T \in \mathrm{End}(V)$ is trigonalizable if there exists a basis $\mathcal{B}$ for V such that $[T]_{\mathcal{B}}^{\mathcal{B}}$ is upper triangular, i.e.,

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} \lambda_1 & \dots & * \\ 0 & \ddots & \vdots \\ 0 & \dots & \lambda_n \end{pmatrix} \quad (n = \dim V). \tag{21.13}$$

Theorem 21.c.4 (*Trigonalizability Condition*): Let V be a finite-dimensional vector space over K and $T \in \mathrm{End}(V)$. Then T is trigonalizable if and only if $P_T(x)$ splits as a product of linear factors, i.e., $P_T(x) = (\lambda_1 - x)^{\ell_1} \dots (\lambda_k - x)^{\ell_k}$.

Short Digression

Theorem 21.c.5 (*Fundamental Theorem of Algebra*): Every polynomial $p(x) \in \mathbb{C}[x]$ with complex coefficients splits as a product of linear factors and a constant, i.e., $p(x) = C \cdot (x - a_1)^{n_1} \dots (x - a_r)^{n_r}$, with $C \in \mathbb{C}$, $r \geq 0$, and $a_i \in \mathbb{C}$.

Remark: The proof is typically covered in an analysis course or complex analysis in the third semester.

Corollary 21.c.6 (*Trigonalizability over* $\mathbb{C}$): Let V be a finite-dimensional vector space over $\mathbb{C}$. Then every $T \in \mathrm{End}(V)$ is trigonalizable.

Proof of the Theorem on Trigonalizability

Proof: $\Longrightarrow$: If there exists a basis $\mathcal{B}$ such that $[T]_{\mathcal{B}}^{\mathcal{B}}$ is upper triangular as in (21.13), then:

$$P_T(x) = \det([T]_{\mathcal{B}}^{\mathcal{B}} - x \cdot I_n) = \det \begin{pmatrix} \lambda_1 - x & \dots & * \\ 0 & \ddots & \vdots \\ 0 & \dots & \lambda_n - x \end{pmatrix} = (\lambda_1 - x) \dots (\lambda_n - x).$$

$\Longleftarrow$: We will use induction on $n = \dim V$.

Base case $n = 1$: In this case, V is 1-dimensional and $[T]_{\mathcal{B}}^{\mathcal{B}} = (\,\lambda\,)$, so T is even diagonalizable.

Inductive step: Let $n \geq 1$. Assume that the theorem holds for all vector spaces of dimension $\leq n$ and every endomorphism of such a space. Let V be an $(n+1)$-dimensional vector space and $T \in \mathrm{End}(V)$ such that $P_T(x)$ splits as a product of linear factors.

Let λ_1 be a zero of $P_T(x)$. This implies that there exists an eigenvector $u_1 \in \ker(T - \lambda_1 \cdot \mathrm{id}_V)$. Extend u_1 to a basis $\mathcal{B}' = (u_1, u_2, \dots, u_{n+1})$ of V. Then the matrix of T in basis $\mathcal{B}'$ has the block form:

$$[T]_{\mathcal{B}'}^{\mathcal{B}'} = \left(\begin{array}{c|c} \lambda_1 & * \cdots * \\ \hline 0 & \\ \vdots & A \\ 0 & \end{array}\right) =: M \tag{21.14}$$

where A is an $n \times n$ matrix. By block determinant properties, we have $P_T(x) = (\lambda_1 - x) \cdot P_A(x)$. Since $P_T(x)$ splits as a product of linear factors, so does $P_A(x)$.

Consider the linear map $T_A : K^n \to K^n$ defined by $T_A(u) = A \cdot u$. By the induction hypothesis, there exists a basis $\mathcal{C}$ of K^n such that:

$$[T_A]_{\mathcal{C}}^{\mathcal{C}} = \begin{pmatrix} \lambda_2 & \dots & * \\ 0 & \ddots & \vdots \\ 0 & \dots & \lambda_{n+1} \end{pmatrix}. \tag{21.15}$$

Denote by $\mathcal{E}$ the standard basis of K^n. We know that $[T_A]_{\mathcal{C}}^{\mathcal{C}} = [\mathrm{id}]_{\mathcal{C}}^{\mathcal{E}} \cdot [T_A]_{\mathcal{E}}^{\mathcal{E}} \cdot [\mathrm{id}]_{\mathcal{E}}^{\mathcal{C}} = Q^{-1}AQ$, where $Q = [\mathrm{id}]_{\mathcal{C}}^{\mathcal{E}}$. Consequently, $Q^{-1}AQ = \begin{pmatrix} \lambda_2 & \dots & * \\ 0 & \ddots & \vdots \\ 0 & \dots & \lambda_{n+1} \end{pmatrix}$.

Now define the $(n+1) \times (n+1)$ block matrix:

$$\underline{P} := \left(\begin{array}{c|c} 1 & 0 \dots 0 \\ \hline 0 & \\ \vdots & Q \\ 0 & \end{array}\right) \in M_{(n+1)\times(n+1)}(K). \tag{21.16}$$

Claim: $\underline{P}$ is invertible with $\underline{P}^{-1} = \left(\begin{array}{c|c} 1 & 0 \dots 0 \\ \hline 0 & \\ \vdots & Q^{-1} \\ 0 & \end{array}\right)$, and $\underline{P}^{-1}M\underline{P} = \left(\begin{array}{c|c} \lambda_1 & * \cdots * \\ \hline 0 & \\ \vdots & Q^{-1}AQ \\ 0 & \end{array}\right)$.

Proof of Claim: This follows from direct block matrix multiplication:

$$\underline{P} \cdot \underline{P}^{-1} = \left(\begin{array}{c|c} 1 & 0 \\ \hline 0 & Q \end{array}\right) \left(\begin{array}{c|c} 1 & 0 \\ \hline 0 & Q^{-1} \end{array}\right) = \left(\begin{array}{c|c} 1 & 0 \\ \hline 0 & I_n \end{array}\right)$$

$$\underline{P}^{-1}M\underline{P} = \left(\begin{array}{c|c} 1 & 0 \\ \hline 0 & Q^{-1} \end{array}\right) \left(\begin{array}{c|c} \lambda_1 & * \\ \hline 0 & A \end{array}\right) \left(\begin{array}{c|c} 1 & 0 \\ \hline 0 & Q \end{array}\right) = \left(\begin{array}{c|c} \lambda_1 & * \\ \hline 0 & Q^{-1}AQ \end{array}\right)$$

Since $Q^{-1}AQ$ is upper triangular with diagonal entries $\lambda_2, \dots, \lambda_{n+1}$, the entire matrix $\underline{P}^{-1}M\underline{P}$ is upper triangular. This proves the claim. ■

We continue with the proof of the Theorem. Recall we have the basis $\mathcal{B}' = (u_1, u_2, \dots, u_{n+1})$, where $[T]_{\mathcal{B}'}^{\mathcal{B}'} = M$.

Claim: There exists a basis $\mathcal{B}$ for V such that $[\mathrm{id}]_{\mathcal{B}'}^{\mathcal{B}} = \underline{P}$.

Proof of Claim: Write the entries of $\underline{P}$ as P_{ij}. Define $m = n+1$ and set vectors $v_1, \dots, v_m$ as linear combinations of $\mathcal{B}'$ using the columns of $\underline{P}$:

$$v_k := \sum_{j=1}^{m} P_{jk} u_j \quad \text{for } k = 1, \dots, m.$$

The set $\mathcal{B} := (v_1, \dots, v_m)$ is a basis for V. To see this, assume $c_1v_1 + \dots + c_mv_m = 0$. Substituting the definitions of v_k and using the linear independence of $(u_1, \dots, u_m)$ yields the matrix equation $\underline{P} \cdot (c_1, \dots, c_m)^T = 0$. Because $\underline{P}$ is invertible, $(c_1, \dots, c_m)^T = 0$, proving linear independence. ■

We now have the basis $\mathcal{B}$. Computing the matrix of T in this new basis yields:

$$[T]_{\mathcal{B}}^{\mathcal{B}} = [\mathrm{id}]_{\mathcal{B}}^{\mathcal{B}'} \cdot [T]_{\mathcal{B}'}^{\mathcal{B}'} \cdot [\mathrm{id}]_{\mathcal{B}'}^{\mathcal{B}} = \underline{P}^{-1} \cdot M \cdot \underline{P}. \tag{21.17}$$

As proven above, $\underline{P}^{-1}M\underline{P}$ is strictly upper triangular. Therefore, T is trigonalizable. The induction is now complete. ■

An Alternative (And More Elegant) Proof

Proof: $\Longleftarrow$: We will use induction on $n = \dim V$.

Define the 1-dimensional subspace $U := \mathrm{Sp}(v_1)$ and consider the quotient space V/U. Since v_1 is an eigenvector, $T(U) \subseteq U$. This invariance property allows us to obtain a well-defined induced endomorphism on the quotient space, $\overline{T} : V/U \to V/U$, defined by $\overline{T}([v]) := [Tv]$.

Extend v_1 to a basis $\mathcal{B}_1 = (v_1, w_2, \ldots, w_{n+1})$ of V. The equivalence classes $[w_2], \ldots, [w_{n+1}]$ form a basis for V/U. In this basis, $\overline{T}$ is represented by an $n \times n$ matrix A, and the full transformation T in basis $\mathcal{B}_1$ takes the block form:

$$[T]_{\mathcal{B}_1}^{\mathcal{B}_1} = \left(\begin{array}{c|ccc} \lambda_1 & * & \ldots & * \\ \hline 0 & & & \\ \vdots & & A & \\ 0 & & & \end{array}\right). \tag{21.18}$$

By the properties of block-triangular determinants, we have $P_T(x) = (\lambda_1 - x) \cdot P_A(x)$. Because $\overline{T}$ has the matrix representation A, we know $P_{\overline{T}}(x) = P_A(x)$, which gives us:

$$P_T(x) = (\lambda_1 - x) \cdot P_{\overline{T}}(x).$$

Since $P_T(x)$ splits into a product of linear factors by our initial assumption, $P_{\overline{T}}(x)$ must also split into linear factors.

Because $\dim(V/U) = n$, we can apply the induction hypothesis to $\overline{T}$. There exists a basis $\mathcal{C} = (z_2, \ldots, z_{n+1})$ for V/U such that its matrix is upper triangular:

$$[\overline{T}]_{\mathcal{C}}^{\mathcal{C}} = \begin{pmatrix} \lambda_2 & \ldots & * \\ 0 & \ddots & \vdots \\ 0 & \ldots & \lambda_{n+1} \end{pmatrix}. \tag{21.19}$$

Now, each element in $\mathcal{C}$ is represented by some vector $v_i \in V$, meaning $z_i = [v_i]$. (Such a v_i is not unique, but we pick one specific representative for every i). Define the list of vectors $\mathcal{B} = (v_1, v_2, \ldots, v_{n+1})$. We claim that $\mathcal{B}$ is a basis for V.

To show this, we check for linear independence. Assume:

$$\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_{n+1} v_{n+1} = 0.$$

Applying the quotient map to this equation (where $[v_1] = 0$), we get:

$$\alpha_2 [v_2] + \cdots + \alpha_{n+1} [v_{n+1}] = 0.$$

This implies that $\alpha_2 z_2 + \cdots + \alpha_{n+1} z_{n+1} = 0$. Since the z_i vectors form a basis for V/U, they are linearly independent, which forces $\alpha_2 = \cdots = \alpha_{n+1} = 0$. Plugging this back into our original sum leaves $\alpha_1 v_1 = 0$. Since $v_1 \neq 0$, we must have $\alpha_1 = 0$. Because the $n+1$ vectors are linearly independent and $\dim V = n+1$, $\mathcal{B}$ is indeed a basis for V.

Finally, we claim that $[T]_{\mathcal{B}}^{\mathcal{B}}$ is upper triangular. For the first column, we already know $T(v_1) = \lambda_1 v_1$. For the remaining columns ($2 \leq i \leq n+1$), we read from the upper-triangular matrix (21.19) that:

$$[\overline{T} z_i]_{\mathcal{C}} = \alpha_{2i} z_2 + \cdots + \alpha_{i-1,i} z_{i-1} + \lambda_i z_i$$

for some scalars $\alpha \in K$. Translating this back to the equivalence classes:

$$[\overline{T} v_i] = \alpha_{2i} [v_2] + \cdots + \alpha_{i-1,i} [v_{i-1}] + \lambda_i [v_i].$$

This implies that the difference between Tv_i and the linear combination of the v's must belong to the kernel of the quotient map, which is U:

$$Tv_i = \alpha_{2i} v_2 + \cdots + \alpha_{i-1,i} v_{i-1} + \lambda_i v_i + u \quad \text{for some } u \in U.$$

Since $U = \mathrm{Sp}(v_1)$, we can write $u = \beta_i v_1$ for some scalar β_i. Thus,

$$Tv_i = \beta_i v_1 + \alpha_{2i} v_2 + \cdots + \alpha_{i-1,i} v_{i-1} + \lambda_i v_i.$$

Because Tv_i only requires basis vectors up to index i, $[T]_{\mathcal{B}}^{\mathcal{B}}$ is upper triangular:

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} \lambda_1 & \beta_2 & \dots & \beta_{n+1} \\ 0 & \lambda_2 & \dots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \lambda_{n+1} \end{pmatrix}.$$

This completes the inductive step, showing that T is trigonalizable. ■

The Minimal Polynomial & The Cayley-Hamilton Theorem

(Section 21.d)

A powerful tool in linear algebra is the ability to evaluate polynomials not just at scalar numbers, but at matrices and endomorphisms. Let $A \in M_{n\times n}(K)$. For any integer $k \geq 1$, the matrix power $A^k = A \cdot \dots \cdot A$ is well-defined. Therefore, for any formal polynomial $g(x) = a_d x^d + a_{d-1}x^{d-1} + \dots + a_1 x + a_0 \in K[x]$, we can define the matrix evaluation $g(A) \in M_{n\times n}(K)$ naturally as:

$$g(A) := a_d A^d + a_{d-1}A^{d-1} + \dots + a_1 A + a_0 I_n.$$

A natural question arises: can we find a polynomial that completely "kills" or annihilates a given matrix?

Claim (*Existence of an Annihilating Polynomial*): For every matrix $A \in M_{n\times n}(K)$, there exists a non-zero polynomial $g \in K[x]$ such that $g(A) = 0$.

Proof: Consider the sequence of successive matrix powers. Since the vector space $M_{n\times n}(K)$ has dimension n^2, we can choose an integer $r \geq n^2$ to construct a sequence of length $r+1$:

$$\underbrace{I_n,\ A,\ A^2,\ A^3,\ \dots,\ A^r}_{r+1 \text{ elements in a space of dimension } n^2} \tag{21.20}$$

Because the number of elements strictly exceeds the dimension of the ambient space, the matrices in (21.20) cannot be linearly independent. This implies there exist scalars $a_0, a_1, \dots, a_r \in K$, not all zero, such that:

$$a_0 I_n + a_1 A + a_2 A^2 + \dots + a_r A^r = 0.$$

By setting $g(x) := a_r x^r + \dots + a_1 x + a_0$, we obtain the desired non-zero annihilating polynomial. ■

A similar logic applies seamlessly to any abstract operator $T \in \mathrm{End}(V)$, where V is a finite-dimensional vector space over K, defining $g(T) := a_d T^d + \dots + a_1 T + a_0 \,\mathrm{id}_V$. Since annihilating polynomials exist, it is mathematically fruitful to look for the "simplest" one, which acts as a sort of algebraic DNA for the operator.

Definition 21.d.1 (*The Minimal Polynomial*): Let V be a vector space over K and $T \in \mathrm{End}(V)$. A **minimal polynomial** for T is a non-zero, monic polynomial $m_T(x) \in K[x]$ of the minimal possible degree such that $m_T(T) = 0$.

While the minimal polynomial guarantees that an operator satisfies *some* polynomial equation, one of the most celebrated theorems in mathematics provides a canonical polynomial that always works: the characteristic polynomial.

Theorem 21.d.2 (*Cayley-Hamilton*): Let $A \in M_{n\times n}(K)$. Then

(a) $P_A(A) = 0$.

(b) Let V be a finite-dimensional vector space over K, and $T \in \mathrm{End}(V)$. Then $P_T(T) = 0$.

Exercise: Let $A \in M_{n\times n}(K)$ be an invertible matrix. Use Theorem 21.d.2 to prove that the inverse matrix A^{-1} can be expressed as a polynomial in A of degree at most $n-1$.

Proof of Cayley-Hamilton Theorem (Analytic Approach)

Before we dive into a purely algebraic mechanism, it is highly instructive to view this theorem through the lens of topology. The core idea is elegant: the theorem is trivially true for diagonal matrices. If we can show that diagonalizable matrices are "dense" (meaning almost all matrices are diagonalizable), we can use the continuity of polynomials to force the theorem to hold for *every* matrix.

First, we note that it is enough to prove part (a) (for matrices). Suppose we know that (a) is true, and let $T \in \mathrm{End}(V)$. Pick a basis $\mathcal{B}$ for V, and assign $A := [T]_{\mathcal{B}}^{\mathcal{B}} \in M_{n\times n}(K)$.

Claim (*Matrix Representation of Polynomials***):** $[P_T(T)]_{\mathcal{B}}^{\mathcal{B}} = P_A(A)$.

Proof of Claim: The map assigning an endomorphism to its matrix representation is an algebra isomorphism. Thus, $[R+S]_{\mathcal{B}}^{\mathcal{B}} = [R]_{\mathcal{B}}^{\mathcal{B}} + [S]_{\mathcal{B}}^{\mathcal{B}}$, and $[R \circ S]_{\mathcal{B}}^{\mathcal{B}} = [R]_{\mathcal{B}}^{\mathcal{B}} \cdot [S]_{\mathcal{B}}^{\mathcal{B}}$. Since $P_T(x) = P_A(x)$ by definition, we write $P_A(x) = c_n x^n + \cdots + c_0$. Then:

$$\begin{aligned}
[P_T(T)]_{\mathcal{B}}^{\mathcal{B}} &= [c_n T^n + \cdots + c_1 T + c_0 \,\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}} \\
&= c_n [T^n]_{\mathcal{B}}^{\mathcal{B}} + \cdots + c_1 [T]_{\mathcal{B}}^{\mathcal{B}} + c_0 I_n \\
&= c_n A^n + \cdots + c_1 A + c_0 I_n = P_A(A).
\end{aligned}$$

■

If Cayley-Hamilton holds for matrices, then $P_A(A) = 0$, meaning $[P_T(T)]_{\mathcal{B}}^{\mathcal{B}} = 0$. Since the zero matrix uniquely represents the zero endomorphism, we conclude $P_T(T) = 0$. We now concentrate entirely on proving the theorem for matrices, beginning over the complex field $\mathbb{C}$.

Lemma 21.d.3 (*Cayley-Hamilton for Diagonalizable Matrices***):** Suppose $A \in M_{n\times n}(\mathbb{C})$ is diagonalizable. Then $P_A(A) = 0$.

Proof: By assumption, there exists an invertible matrix Q such that $Q^{-1}AQ = \Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$. Since the roots of the characteristic polynomial are exactly the eigenvalues, we know $P_A(\lambda_i) = 0$ for all i. Writing $P_A(x) = \sum c_j x^j$, we substitute A:

$$Q^{-1}P_A(A)Q = \sum c_j Q^{-1}A^j Q = \sum c_j (Q^{-1}AQ)^j = \sum c_j \Lambda^j = P_A(\Lambda).$$

Because Λ is diagonal, $P_A(\Lambda)$ is simply the diagonal matrix with entries $P_A(\lambda_1), \ldots, P_A(\lambda_n)$. Since every entry is zero, $Q^{-1}P_A(A)Q = 0$, which immediately implies $P_A(A) = 0$. ■

Lemma 21.d.4 (*Density of Diagonalizable Matrices***):** Let $A \in M_{n\times n}(\mathbb{C})$ be any arbitrary matrix. Then there exists a sequence of matrices $(A_k)_{k=1}^{\infty}$ such that:

(a) Each A_k is diagonalizable.

(b) $\lim_{k\to\infty} A_k = A$.

Proof: Because $\mathbb{C}$ is algebraically closed, every matrix is trigonalizable. Thus, there exists an invertible matrix P such that $U = P^{-1}AP$ is an upper triangular matrix with the eigenvalues $\lambda_1, \ldots, \lambda_n$ on its diagonal.

To ensure diagonalizability, we must prevent repeated eigenvalues. We can fix this by introducing a tiny perturbation. For each integer $k \geq 1$, we define a diagonal matrix of small shifts:

$$E_k = \mathrm{diag}\left(\frac{\epsilon_1}{k}, \frac{\epsilon_2}{k}, \ldots, \frac{\epsilon_n}{k}\right)$$

We carefully choose the constants ϵ_i such that the new diagonal entries $\lambda_i + \frac{\epsilon_i}{k}$ are strictly distinct from one another. We now define our sequence:

$$A_k := P(U + E_k)P^{-1}$$

Because $U + E_k$ is still upper triangular but now possesses exactly n distinct eigenvalues, it is guaranteed to be diagonalizable. Finally, as $k \to \infty$, the perturbation matrix $E_k \to 0$, meaning $\lim_{k\to\infty} A_k = PUP^{-1} = A$. ■

Lemma 21.d.5 (*Cayley-Hamilton over* $\mathbb{C}$): Cayley-Hamilton holds for every matrix $A \in M_{n\times n}(\mathbb{C})$.

Proof: By our construction in Lemma 21.d.4, we can find a sequence of diagonalizable matrices $A_k \to A$. By Lemma 21.d.3, the theorem holds for all of them: $P_{A_k}(A_k) = 0$.

Notice that the mapping $\Phi : M_{n\times n}(\mathbb{C}) \to M_{n\times n}(\mathbb{C})$ defined by $\Phi(M) = P_M(M)$ is built entirely from taking determinants, adding matrices, and multiplying matrices. Therefore, Φ is a continuous polynomial function of the matrix entries. Because continuous functions commute with limits, we have:

$$P_A(A) = \Phi(A) = \Phi\left(\lim_{k\to\infty} A_k\right) = \lim_{k\to\infty} \Phi(A_k) = \lim_{k\to\infty} 0 = 0.$$

■

To generalize this result from $\mathbb{C}$ to any arbitrary field K, we rely on a formal algebraic identity regarding multi-variable polynomials.

Lemma 21.d.6 (*Formal Zero of Multi-Variable Polynomials*): Suppose $f(x_1, \dots, x_r) \in \mathbb{R}[x_1, \dots, x_r]$ satisfies $f(a_1, \dots, a_r) = 0$ for all $(a_1, \dots, a_r) \in \mathbb{R}^r$. Then $f \equiv 0$ as a formal polynomial (all coefficients are 0).

Proof: Write $f(x_1, \dots, x_r) = \sum_{\mathbf{i}\in\mathcal{I}} c_{\mathbf{i}} x_1^{i_1} \dots x_r^{i_r}$. Because f is zero everywhere on $\mathbb{R}^r$, all of its partial derivatives evaluated at the origin $(0, \dots, 0)$ must be zero. Taking the appropriate mixed partial derivatives isolates the coefficient $c_{\mathbf{i}}$. Thus, $c_{\mathbf{i}} = 0$ for all $\mathbf{i}$. ■

Analytic Proof of Cayley-Hamilton over every field K: For an $n \times n$ matrix A, the entries of $P_A(A)$ are given by multi-variable polynomials in the n^2 variables $A_{11}, A_{12}, \dots, A_{nn}$. These polynomials have integer (and hence real) coefficients. Let the polynomial representing the (i, j)-th entry be $q_{ij}(A_{11}, \dots, A_{nn})$.

Since Cayley-Hamilton holds for $\mathbb{C}$ (and thus for any matrix in $\mathbb{R}$), we know $q_{ij}(A_{11}, \dots) = 0$ for all evaluations in $\mathbb{R}^{n^2}$. By Lemma 21.d.6, $q_{ij} \equiv 0$ as formal polynomials. Because they are formally the zero polynomial, evaluating them over *any* field K will always yield 0. Thus, $P_A(A) = 0$ for any matrix over any field. ■

A Purely Algebraic Proof of Cayley-Hamilton Thm

While the analytic proof is mathematically beautiful, it relies on topological machinery (limits and continuity) that feels slightly alien in pure algebra. Let us now construct a purely structural, algebraic proof using invariant subspaces.

Definition 21.d.7 (*T-Cyclic Subspace*): Let V be a vector space over K and $T \in \mathrm{End}(V)$. Let $v \in V$. We define the **T-cyclic subspace** generated by v as the span of the orbit of v under T:

$$\langle v\rangle_T := \mathrm{Sp}\{T^k v \mid k \in \mathbb{Z}_{\geq 0}\} = \mathrm{Sp}(v, Tv, T^2v, \dots).$$

Lemma 21.d.8 (*Basis and Companion Matrix*): Suppose that $v, Tv, \dots, T^{d-1}v$ $(d \geq 1)$ are linearly independent, and $T^d v = c_0 v + c_1 Tv + \dots + c_{d-1}T^{d-1}v$. Then:

(a) $\mathcal{B} = (v, Tv, \dots, T^{d-1}v)$ forms a basis for $\langle v\rangle_T$.

(b) Let $S = T|_{\langle v\rangle_T}$. Then $[S]_{\mathcal{B}}^{\mathcal{B}}$ is a companion matrix:

$$[S]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 0 & 0 & \dots & 0 & c_0 \\ 1 & 0 & \dots & 0 & c_1 \\ 0 & 1 & \dots & 0 & c_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & c_{d-1} \end{pmatrix}.$$

Proof: (a) Let $U = \mathrm{Sp}(\mathcal{B})$. Applying T to any basis vector $T^k v$ yields either the next basis element $T^{k+1}v \in U$, or (in the case of $k = d-1$) it yields $T^d v \in U$ due to the given linear combination. Thus $T(U) \subseteq U$. Since $\langle v\rangle_T$ is by definition the smallest T-invariant subspace containing v, we must have $U = \langle v\rangle_T$. Since the set is linearly independent and spans the space, it is a basis.

(b) This follows directly from the definition of the matrix representation; each vector maps to the next standard basis vector, creating the 1s on the subdiagonal, and the last vector maps to the prescribed linear combination, populating the final column. ■

Lemma 21.d.9 (*Characteristic Polynomial of a Companion Matrix*): Under the assumptions of Lemma 21.d.8, the characteristic polynomial of the restriction S is given by:

$$P_S(x) = (-1)^d(x^d - c_{d-1}x^{d-1} - \cdots - c_1x - c_0).$$

Proof: We compute $\det([S]_{\mathcal{B}}^{\mathcal{B}} - xI_d)$ using cofactor expansion along the first row:

$$P_S(x) = \begin{vmatrix} -x & 0 & \dots & 0 & c_0 \\ 1 & -x & \dots & 0 & c_1 \\ 0 & 1 & \dots & 0 & c_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & c_{d-1} - x \end{vmatrix}$$

Expanding across the first row, we get two non-zero terms:

$$P_S(x) = (-x)\det(M_{11}) + (-1)^{1+d}c_0\det(M_{1d})$$

Notice that M_{1d} is a lower triangular matrix with entirely 1s on its main diagonal. Thus, $\det(M_{1d}) = 1$. The matrix M_{11} is exactly the $(d-1)\times(d-1)$ matrix representing the characteristic polynomial of the smaller companion matrix. By induction on d, assuming the formula holds for $\det(M_{11})$, multiplying it by $-x$ shifts the degree of all terms up by one. Adding the $(-1)^{d+1}c_0$ term gives the exact formula claimed. ■

Algebraic Proof of Cayley-Hamilton: Let V be finite-dimensional, and $T \in \mathrm{End}(V)$. We want to show $P_T(T) = 0$, meaning $P_T(T)v = 0$ for all $v \in V$.

Let $v \in V$, $v \neq 0$. Consider the cyclic subspace $\langle v \rangle_T$. Take the maximal integer $d \geq 1$ such that $v, Tv, \dots, T^{d-1}v$ are linearly independent. Then $T^dv = c_0v + \cdots + c_{d-1}T^{d-1}v$. Extend this basis $\mathcal{B}$ to a full basis $\mathcal{C} = (v, \dots, T^{d-1}v, w_1, \dots, w_{n-d})$ for V. The matrix representation is block upper-triangular:

$$[T]_{\mathcal{C}}^{\mathcal{C}} = \begin{array}{|c|c|} \hline [S]_{\mathcal{B}}^{\mathcal{B}} & * \\ \hline 0 & [C] \\ \hline \end{array}$$

Because the determinant of a block-triangular matrix is the product of the determinants of its diagonal blocks, we have $P_T(x) = P_S(x) \cdot P_C(x)$. Consider the endomorphism $R := P_T(T) \in \mathrm{End}(V)$. Since polynomials evaluated at T commute with one another, we can factor R:

$$R = P_C(T) \circ P_S(T).$$

Therefore, $Rv = P_C(T)\Big(P_S(T)v\Big)$. However, applying Lemma 21.d.9 directly to v:

$$P_S(T)v = (-1)^d(T^dv - c_{d-1}T^{d-1}v - \cdots - c_0v).$$

By the very definition of our constants c_i, this evaluates exactly to 0. Hence $Rv = P_C(T)(0) = 0$. Since $v \neq 0$ was chosen completely arbitrarily, $P_T(T)$ evaluates to zero on every vector in V, meaning $P_T(T) = 0$. ■

Remark (*The Lazy Proof*): A completely incorrect "proof" often presented by students is: $P_A(A) = \det(A - A \cdot I) = \det(0) = 0$. This is fundamentally invalid. The expression $P_A(x) = \det(A - xI)$ evaluates to a polynomial with scalar coefficients. Substituting matrices into a scalar expression *before* expanding the determinant leads to a severe category error. One must compute the scalar polynomial first, and only then evaluate it at the matrix A.